

Unit 6 - Dilations and Similarity

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The following B.E.S.T. Standards will be covered in this unit:

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle and Hypotenuse-Leg.

- *Clarification 1: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flowchart proofs or informal proofs.*
- *Clarification 2: Instruction focuses on helping a student choose a method they can use reliably.*

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

- *Clarification 1: Instruction includes demonstrating that two-dimensional figures are congruent or similar based on given information.*

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

- *Clarification 1: Instruction includes the connection of transformations to functions that take points in the plane as inputs and give other points in the plane as outputs.*
- *Clarification 2: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.*
- *Clarification 3: Within the Geometry course, rotations are limited to 90° , 180° and 270° counterclockwise or clockwise about the center of rotation, and the centers of rotations and dilations are limited to the origin or a point on the figure.*

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

- *Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.*
- *Clarification 2: Instruction includes recognizing that these transformations preserve angle measure.*

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

- *Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.*
- *Clarification 2: Within the Geometry course, figures are limited to triangles and quadrilaterals and rotations are limited to 90° , 180° and 270° counterclockwise or clockwise about the center of rotation.*
- *Clarification 3: Instruction includes the understanding that when a figure is mapped onto itself using a reflection, it occurs over a line of symmetry.*

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.

- *Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.*
- *Clarification 2: Instruction includes two or more transformations.*

MA.912.GR.2.8 Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar.

- *Clarification 1: Instruction includes showing that the corresponding sides are proportional, and the corresponding angles are congruent.*

MA.912.GR.2.9 (Honors only) Justify the criteria for triangle similarity using the definition of similarity in terms of nonrigid transformations.

MA.912.GR.6.5 (Honors only) Apply transformations to prove that all circles are similar.

MA.912.LT.4.8 (Honors only) Construct proofs, including proofs by contradiction.

- *Clarification 1: Within the Geometry course, proofs are limited to geometric statements within the course.*

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Unit 6, Lesson 1: Dilations



Benchmark

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.



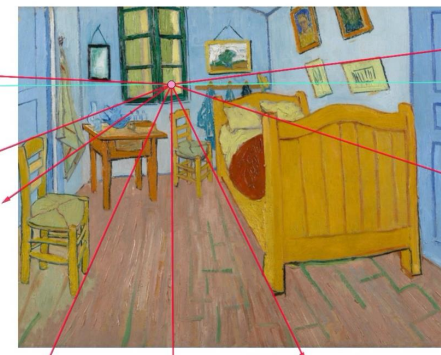
Learning Targets

- ☐ I can write an algebraic description using coordinate notation of a dilation on a coordinate plane.
- ☐ I can describe a dilation by giving the scale factor and center of dilation.



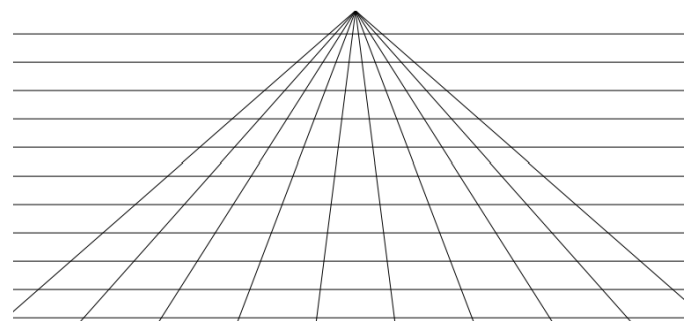
6.1.1 Warm-Up: What Is Your Perspective?

Point perspective is a common method of drawing a three-dimensional object onto a two-dimensional plane. The artist chooses one point from which everything in the drawing is set. It was first discovered during the Renaissance. An



example of a painting that uses point perspective is Vincent van Gogh's *Vincent's Bedroom in Arles*.

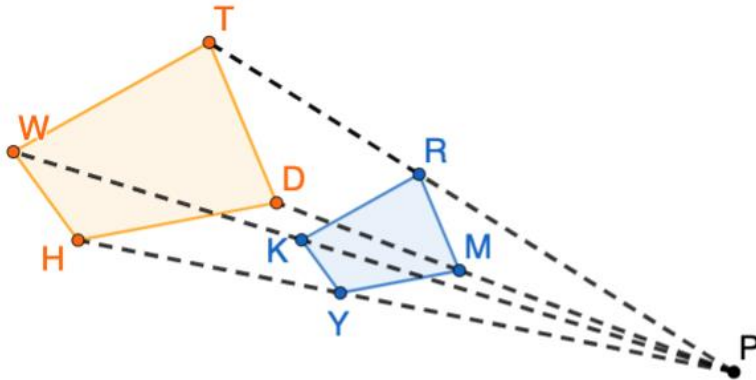
1. Use the grid to sketch a rectangle that is close to the point of perspective and a rectangle that is farther away from the point of perspective.





6.1.2 Refresher: Dilations

1. $KRMY$ was created by dilating $WTDH$. The ratio of WT to KR is $\frac{5}{3}$.



- a. Use the words in the bank below to complete the statements that follow.

center of dilation
pre-image

image
scale factor

Point P is the _____.

$KRMY$ is the _____.

$WTDH$ is the _____.

The _____ is $\frac{3}{5}$.

- b. Use the diagram to determine each.

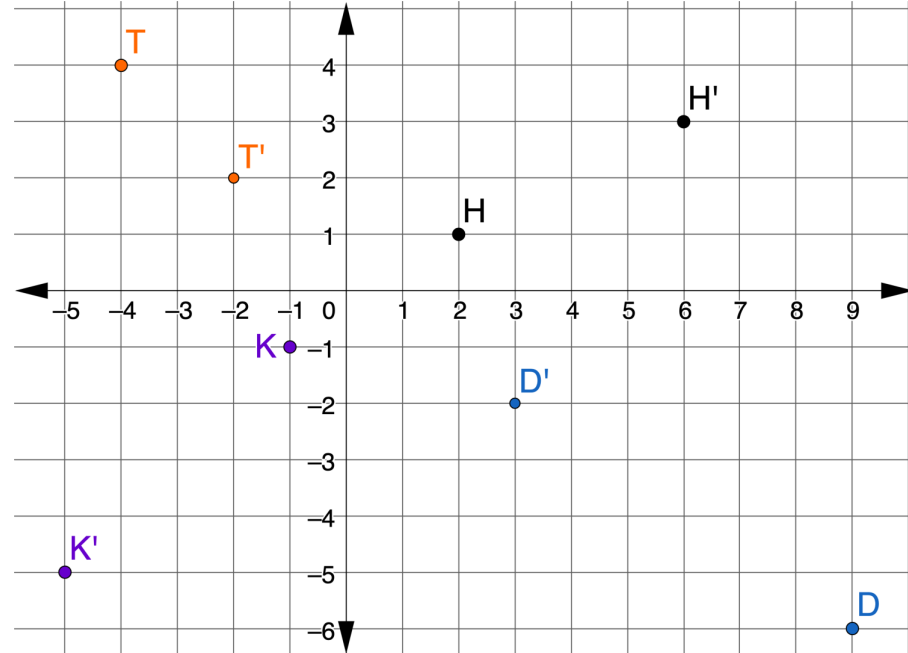
What is the ratio of $\frac{WH}{KY}$?

What is the ratio of $\frac{PR}{RT}$?



6.1.3 Collaboration: Algebraic Description of a Dilation

1. Points D , H , K , and T are shown on the coordinate grid. Each point has been dilated using $(0,0)$ as the center of dilation.



- a. Complete the table.

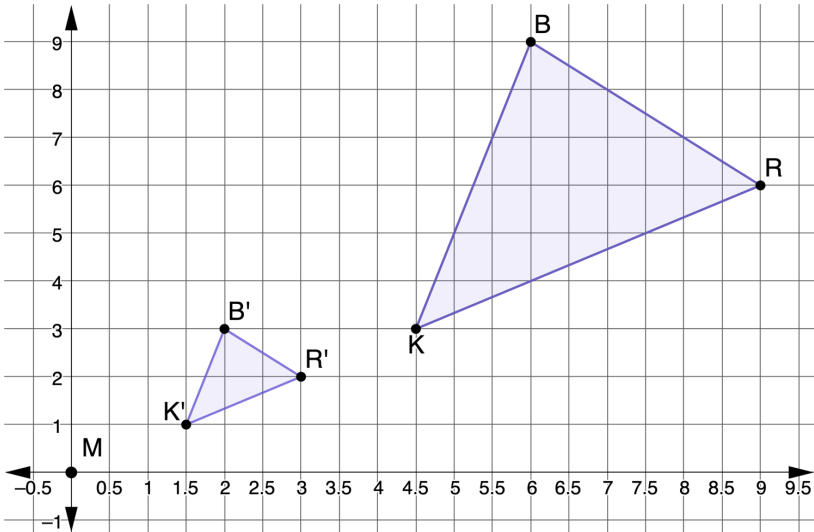
Point	Ordered Pair	Ordered Pair of the Dilation	Scale Factor
D			
H			
K			
T			

In the coordinate plane, the algebraic representation of a dilation centered at the origin is $(x,y) \rightarrow (kx,ky)$, where k is the scale factor.

b. Write the algebraic representation for each point.

Point	Algebraic Representation of Dilation
D	
H	
K	
T	

2. $\Delta R'B'K'$ is the result of dilating ΔRBK using the origin as the center of dilation.



a. What is the scale factor of the dilation?

b. Write the algebraic description of the dilation.

c. Draw a line segment from point M that passes through B' and ends at B .

d. The table shows different relationships of the line segments MB , $B'B$, and MB' .

Line Segment	Slope	Distance
MB'	$\frac{3}{2}$	$\sqrt{2^2 + 3^2} = \sqrt{13}$
$B'B$	$\frac{6}{4}$	$\sqrt{6^2 + 4^2} = \sqrt{52} = 2\sqrt{13}$
MB	$\frac{9}{6}$	$\sqrt{6^2 + 9^2} = \sqrt{117} = 3\sqrt{13}$

e. Discuss with your partner how the scale factor could be determined using slope. Summarize your discussion.

f. Discuss with your partner how the scale factor could be determined using distance.

3. The table shows different relationships among the line segments BR , $B'R'$, KR and $K'R'$.

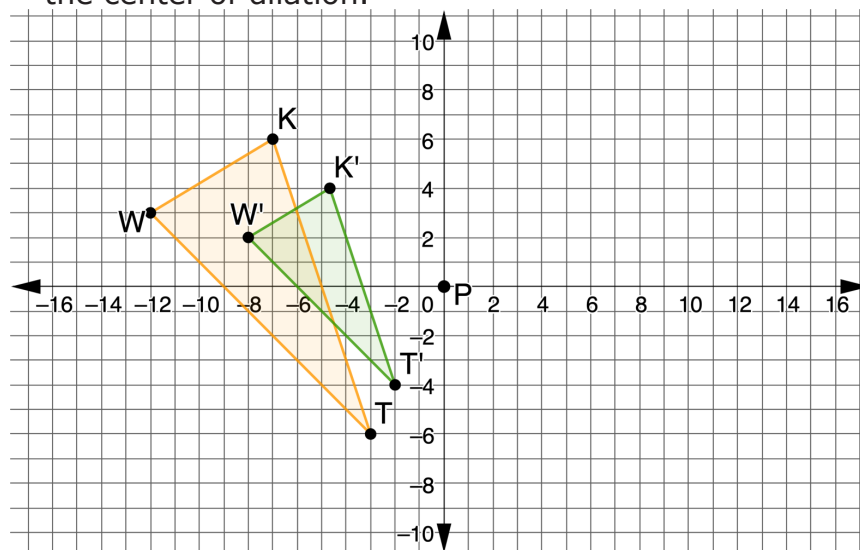
Line Segment	Slope	Distance
BR	$-\frac{3}{3}$	$\sqrt{3^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$
$B'R'$	$-\frac{1}{1}$	$\sqrt{1^2 + 1^2} = \sqrt{2}$
KR	$\frac{3}{4.5}$	$\sqrt{3^2 + 4.5^2} = \sqrt{29.25}$
$K'R'$	$\frac{1}{1.5}$	$\sqrt{1^2 + 1.5^2} = \sqrt{3.25}$

Discuss with your partner how the scale factor is also found in the relationships between the slopes of BR and $B'R'$, and KR and $K'R'$. Write a summary of the relationships.



6.1.4 Guided Instruction: Describing Dilations

- $\Delta K'M'T'$ is the result of dilating ΔKMT using the origin as the center of dilation. The vertices of ΔKMT are located at $K(-8, -6)$, $M(6, 7)$, and $T(0, 4)$. The vertices of $\Delta K'M'T'$ are located at $K'(-12, -9)$, $M'(9, 10.5)$, and $T'(0, 6)$. Write the algebraic description of the dilation.
- Triangles TWK and $T'W'K'$ are shown on the coordinate grid, where $\Delta T'W'K'$ is a dilation of ΔTWK and point P is the center of dilation.

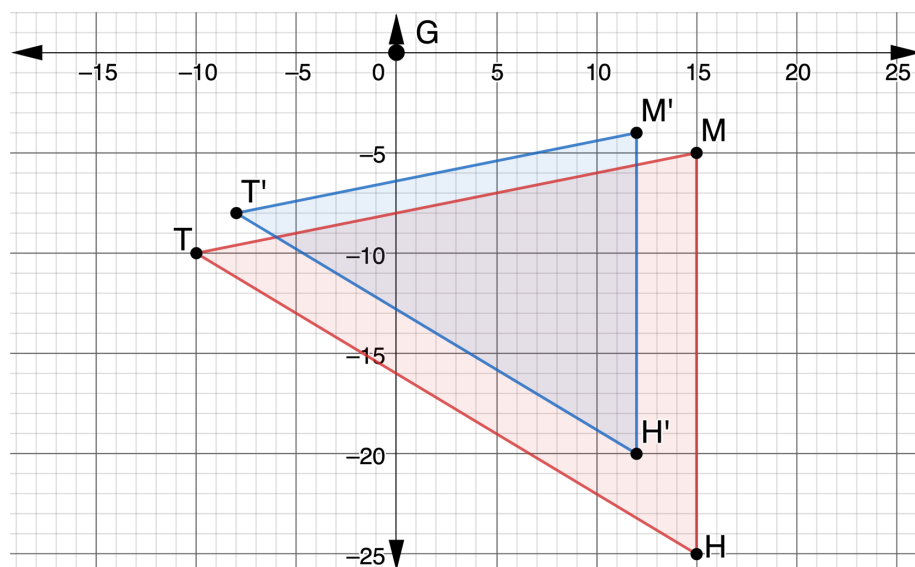


- Show the dilation on the coordinate grid by drawing dotted lines from the center of dilation through each vertex.

b. What is the scale factor of the dilation?

c. Write the algebraic description of the dilation.

3. Triangles THM and $T'H'M'$ are shown on the coordinate grid, where $\Delta T'H'M'$ is a dilation of ΔTHM .



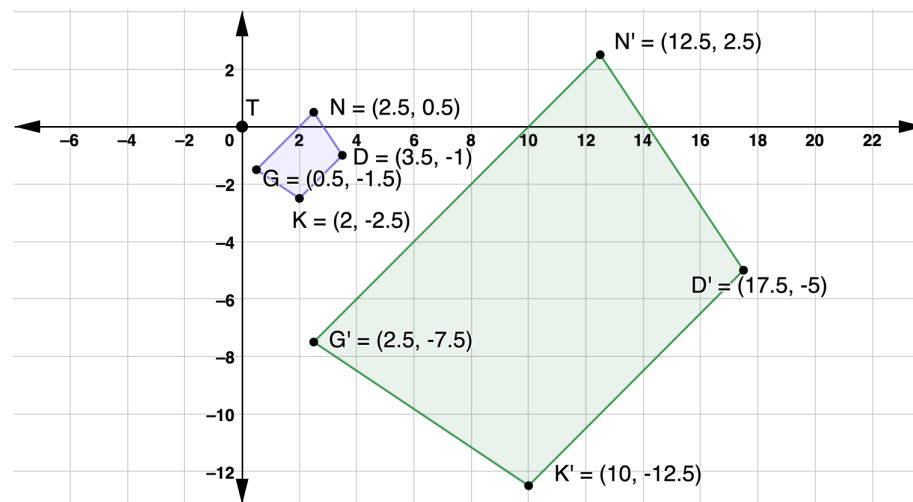
Write a description of the dilation by giving the center of dilation and the scale factor.



6.1.5 Guided Practice: Describing Dilations

1. $\overline{M'T'}$ has coordinates $M'(-16, 20)$ and $T'(4, -8)$, and is the result of the dilation of $\overline{MT}$ centered at the origin. The coordinates of $\overline{MT}$ are $M(-4, 5)$ and $T(1, -2)$. Write the algebraic description of the dilation.

2. $NDKG$ and $N'D'K'G'$ are shown on the coordinate grid, where $N'D'K'G'$ is a dilation of $NDKG$.



Write a description of the dilation by giving the center of dilation and the scale factor.



6.1.6 Wrap-Up: Cool Down

1. Explain how dilations are different from rigid transformations, in your own words.

Unit 6, Lesson 2: Sequences of Transformations That Include Dilations



Benchmarks

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.



Learning Targets

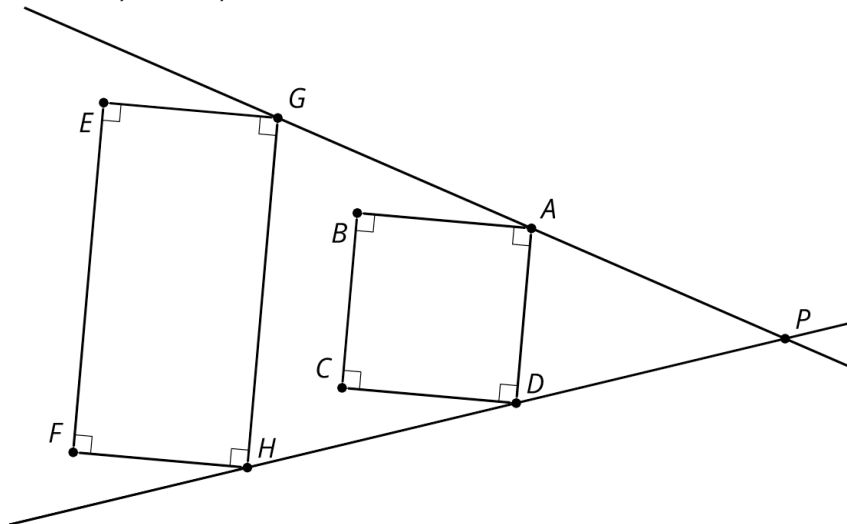
- ☐ I can describe a sequence of transformations that includes a dilation.
- ☐ I can represent a sequence of transformations using coordinate notation with algebraic descriptors.



6.2.1 Warm-Up: Dilation Miscalculation

1. Cassandra tried to determine if the diagram shown represents a dilation.

$\overrightarrow{EGHF}$, $\overrightarrow{BADC}$, $\overrightarrow{PG}$ and $\overrightarrow{PH}$ are shown.



Cassandra listed four characteristics of dilations:

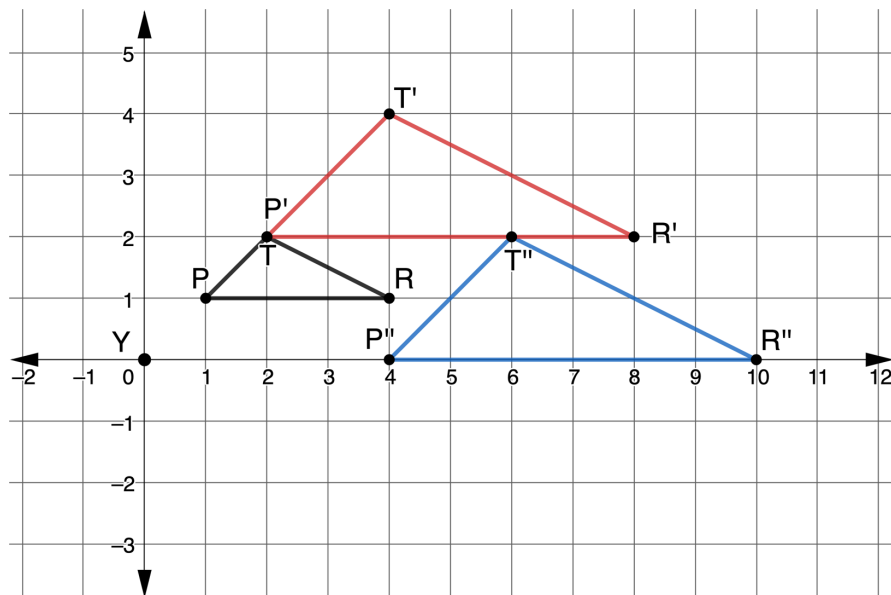
Characteristic	Yes/No
Corresponding angle measures are equal	Yes
Corresponding sides are parallel	Yes
Corresponding vertices are collinear with the point of dilation	Yes
Orientation of the shapes are the same	Yes

- a. Of the characteristics Cassandra listed, which are not true for $\overrightarrow{EGHF}$ and $\overrightarrow{BADC}$?

- b. Explain why the diagram shows that the dilation does not have these characteristics. If necessary, use the diagram to draw.



1. A sequence of transformations of ΔPTR is shown on the coordinate grid. ΔPTR is dilated then translated, where $\Delta P'T'R'$ is the dilation and $\Delta P''T''R''$ is the translation of $\Delta P'T'R'$.



Triangle PTR	
P	$(1, 1)$
T	$(2, 2)$
R	$(4, 1)$

Triangle $P'T'R'$	
P'	$(2, 2)$
T'	$(4, 4)$
R'	$(8, 2)$

Triangle $P''T''R''$	
P''	$(4, 0)$
T''	$(6, 2)$
R''	$(10, 0)$

- a. Complete the description.

ΔPTR was dilated using a scale factor of _____ centered at the origin. The resulting polygon, $\Delta P'T'R'$, was translated _____ to create $\Delta P''T''R''$.

- b. Complete the table.

Transformation	Algebraic Description
Dilation	$(x, y) \rightarrow (\text{___}x, \text{___}y)$
Translation	$(x, y) \rightarrow (x\text{___}, y\text{___})$

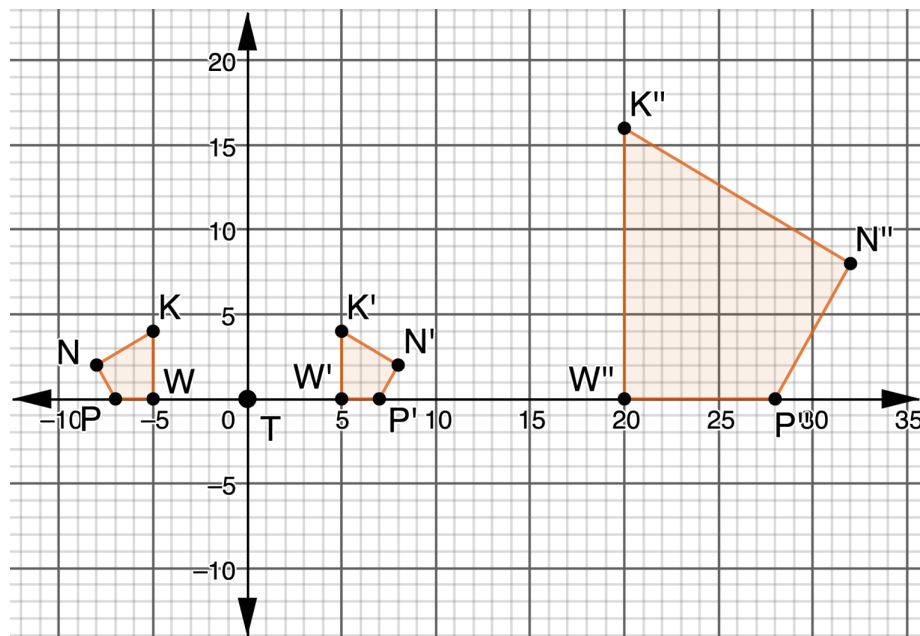
- c. Determine the coordinates of each triangle if $\triangle PTR$ is first translated then dilated.

Pre-image		Translation		Dilation	
Triangle PTR		Triangle $P'T'R'$		Triangle $P''T''R''$	
P	$(1, 1)$	P'		P''	
T	$(2, 2)$	T'		T''	
R	$(4, 1)$	R'		R''	

Some sequences of transformations are commutative, which means the transformations can be applied in any order.

- d. Use the tables to determine if the sequence of transformations applied to ΔPTR is commutative. Work with your partner to write an explanation.

2. $KWPN$, $K'W'P'N'$, and $K''W''P''N''$ are shown on the coordinate grid.



$KWPN$		$K'W'P'N'$		$K''W''P''N''$	
K	$(-5, 4)$	K'	$(5, 4)$	K	$(20, 16)$
W	$(-5, 0)$	W'	$(5, 0)$	W	$(20, 0)$
P	$(-7, 0)$	P'	$(7, 0)$	P	$(28, 0)$
N	$(-8, 2)$	N'	$(8, 2)$	N	$(32, 8)$

- Discuss with your partner the first transformation that occurred. Describe how you determined the first transformation.
- Determine the scale factor in the second transformation.

- Complete the description of the transformations.

$KWPN$ was _____ to create $K'W'P'N'$. $K'W'P'N'$ was dilated using a scale factor of _____ centered at _____ to create $K''W''P''N''$.

- Complete the table.

Transformation	Algebraic Description
	$(x, y) \rightarrow$
	$(x, y) \rightarrow$

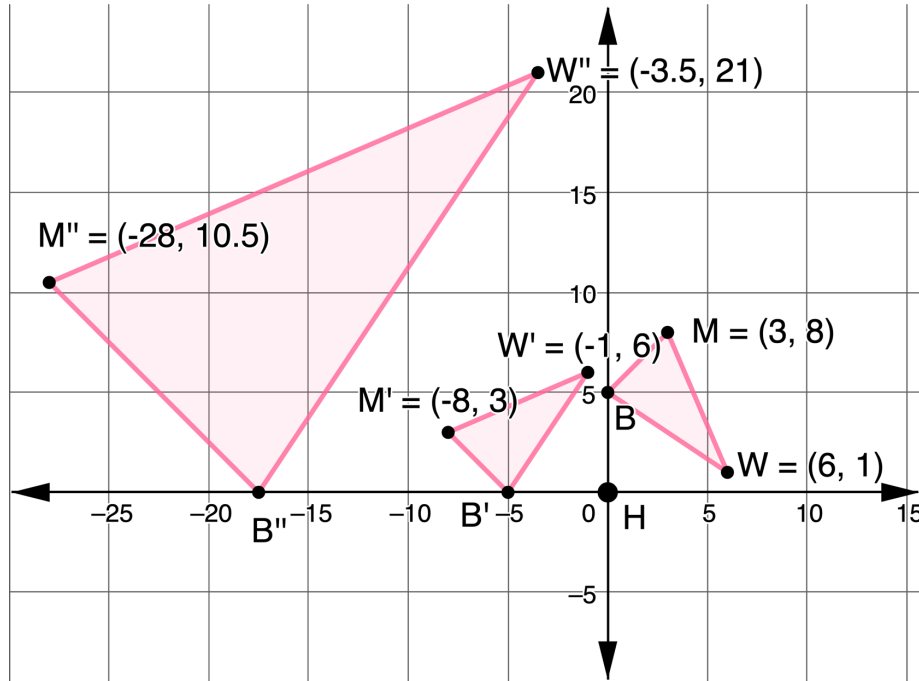
- Work with your partner to determine if the sequence of transformations applied to $KWPN$ is commutative. Complete the table by first applying the dilation, and then the other transformation. Then, write a conclusion based on the tables.

$KWPN$		$K'W'P'N'$		$K''W''P''N''$	
K	$(-5, 4)$	K'		K	
W	$(-5, 0)$	W'		W	
P	$(-7, 0)$	P'		P	
N	$(-8, 2)$	N'		N	



6.2.3 Guided Practice: Mapping a Dilation

- A sequence of transformations of $\triangle MWB$ is shown on the coordinate grid.



- Complete the table of values for each triangle.

Preimage

Dilation

$\triangle MWB$		$\triangle M'W'B'$		$\triangle M''W''B''$	
M	(3, 8)	M'	(-8, 3)	M''	(-28, 10.5)
W	(6, 1)	W'	(-1, 6)	W''	(-3.5, 21)
B		B'		B''	

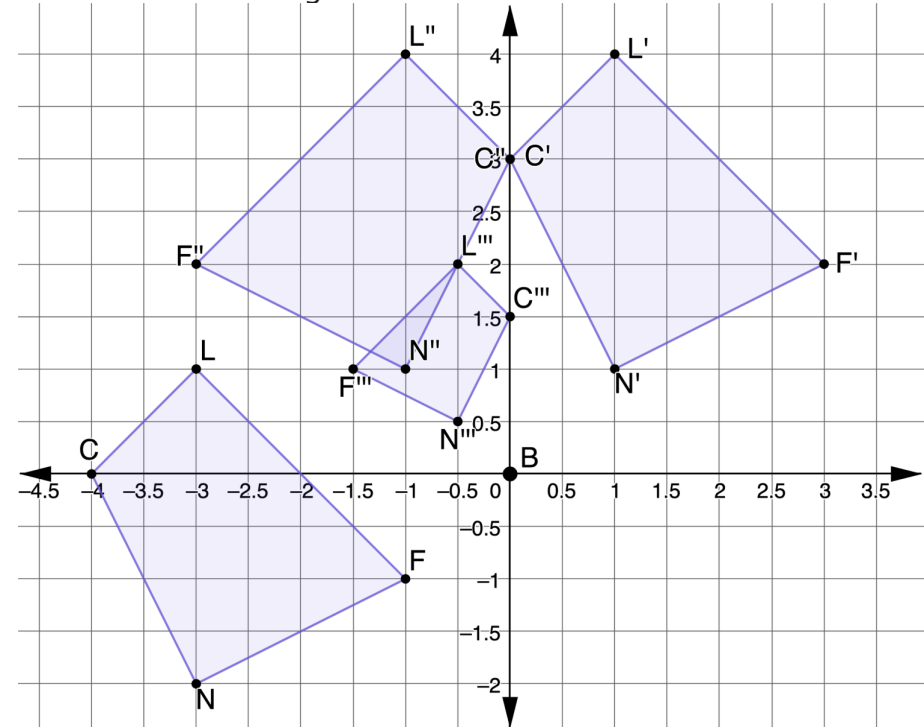
- Complete the table.

Transformation	Algebraic Description
	$(x, y) \rightarrow$
	$(x, y) \rightarrow$



6.2.4 Your Turn!

- $LFNC$, $L'F'N'C'$, $L''F''N''C''$, and $L'''F'''N'''C'''$ are shown on the coordinate grid.



- Describe the sequence of transformations of $LFNC$ that resulted in $L'''F'''N'''C'''$.

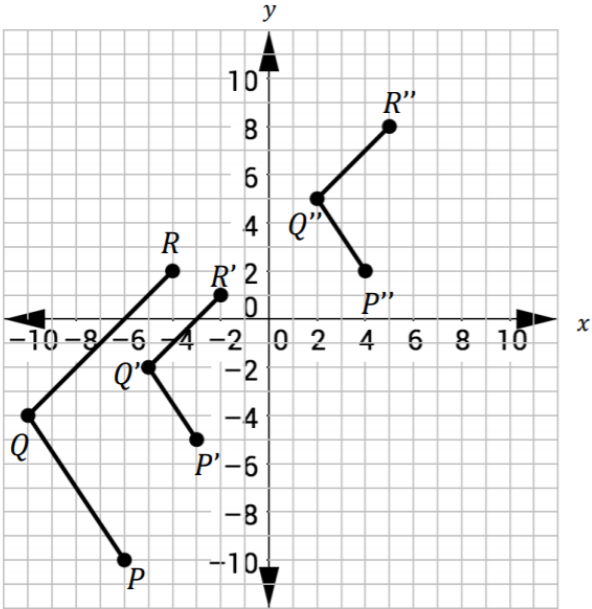
- b. Write the algebraic descriptions of the sequence of transformations.

Transformation	Algebraic Description
	$(x, y) \rightarrow$
	$(x, y) \rightarrow$
	$(x, y) \rightarrow$



6.2.5 Wrap-Up: Ticket Out the Door

1. Points $P(-6, -10)$, $Q(-10, -4)$, and $R(-4, 2)$ are plotted on the coordinate grid to form figure PQR . Two transformations of figure PQR are shown.



Gladys first transformed figure PQR to $P'Q'R'$ using a dilation centered at the origin. She then transformed figure $P'Q'R'$ to $P''Q''R''$. Complete the table.

Transformation	Written Description	Algebraic Description
PQR to $P'Q'R'$		$(x, y) \rightarrow$
$P'Q'R'$ to $P''Q''R''$		$(x, y) \rightarrow$

Unit 6, Lesson 3: Preserving Distance



Benchmarks

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.



Learning Targets

- ☐ I can use a description of a sequence of transformations to determine if the transformation does or does not preserve distance.
- ☐ I can draw a sequence of transformations to determine if the transformation does or does not preserve distance.
- ☐ I can identify transformations that do or do not preserve distance.



6.3.1 Warm-Up: Notice and Wonder

Artist John Hoskins sketched a drawing of Mary Hawtrey, also known as Lady Mary Bankes, in 1821. Henry Pierce Bone then used the sketch to create an enamel miniature painting.

Following the death of her husband, Lady Mary Bankes successfully defended the Corfe Castle in Dorset, England during a siege in 1643. Unfortunately, during a second siege in 1646 the castle was destroyed.



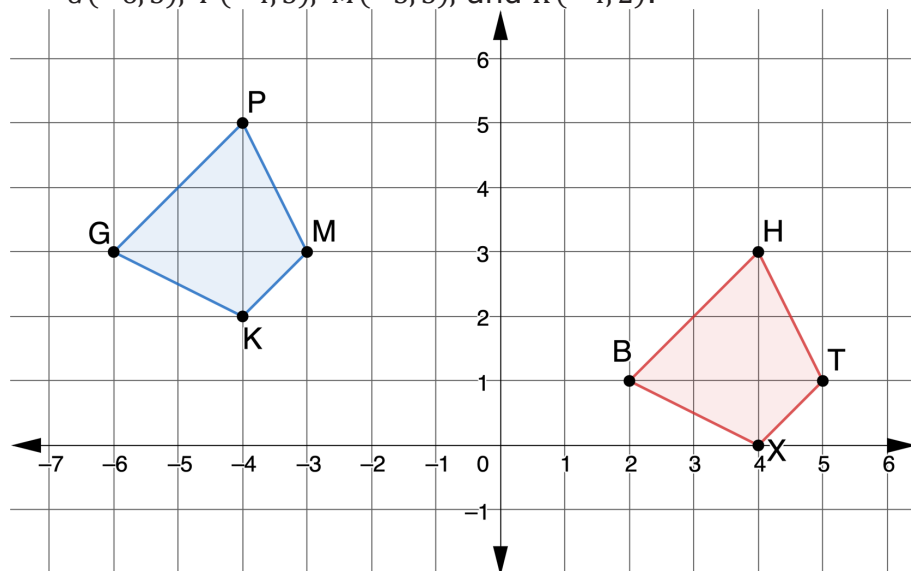
1. What do you notice about the sketch of Lady Mary Bankes?
2. What do you wonder about the sketch of Lady Mary Bankes or her story?



6.3.2 Exploration: Sequence of Transformations

With your partner, answer the following questions.

1. $BHTX$ with vertices $B(2,1)$, $H(4,3)$, $T(5,1)$, and $X(4,0)$ is the result of a transformation of $GPMK$ with vertices $G(-6,3)$, $P(-4,5)$, $M(-3,3)$, and $K(-4,2)$.



- a. What transformation occurred to $GPMK$ that resulted in $BHTX$?
- b. Use patty paper to trace $GPMK$, then place $GPMK$ onto $BHTX$.
- c. What do you notice about the distances between the vertices of corresponding sides?

- d. What can be concluded about the lengths of the corresponding sides after this transformation occurs?

2. The coordinates of $RWTY$ and $R'W'T'Y'$ are shown in the table, where $RWTY$ is the preimage.

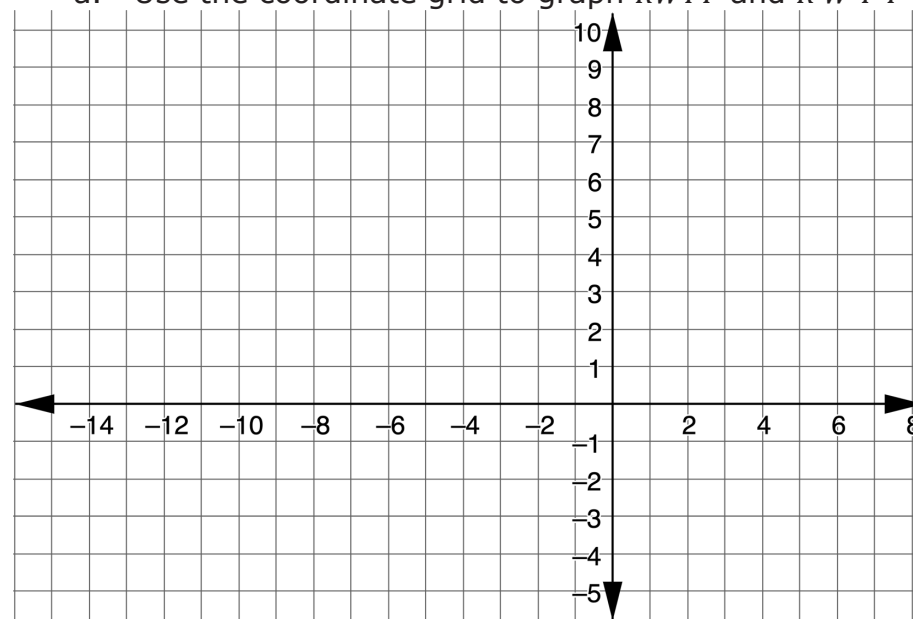
$RWTY$

Vertex	x	y
R	-6	8
W	-4	6
T	2	4
Y	-6	2

$R'W'T'Y'$

Vertex	x	y
R'	-2	-4
W'	0	-2
T'	2	4
Y'	4	-4

- a. Use the coordinate grid to graph $RWTY$ and $R'W'T'Y'$.



- b. What type of transformation was applied to the preimage, $RWTY$?
- c. How could the coordinates of the preimage and image be used to show that the transformation preserved side length?

3. The coordinates of $MPRD$ and $SGTW$ are shown in the table, where $MPRD$ is the preimage.

MPRD

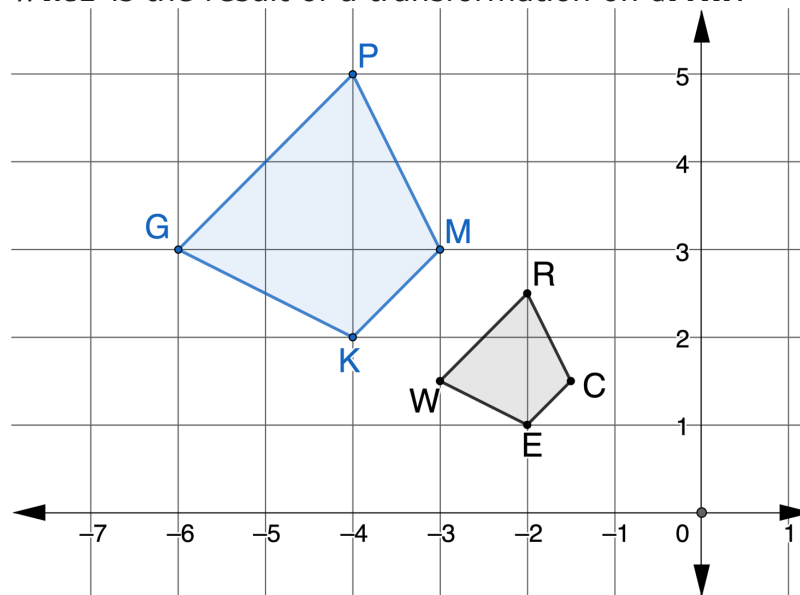
Vertex	x	y
M	8	-7
P	9	-4
R	4	-2
D	1	-5

SGTW

Vertex	x	y
S	8	7
G	9	4
T	4	2
W	1	5

- a. What transformation occurred from the preimage to the image?
- b. Explain how the coordinates of the preimage and image show that the transformation preserved distance.

4. $WRCE$ is the result of a transformation on $GPMK$.



- a. What transformation occurred to $GPMK$ that resulted in $WRCE$?
- b. What can be concluded about the lengths of each corresponding side after this transformation occurs?



6.3.3 Bring It Together: Preserving Distance

1. Complete the statements based on the Exploration.

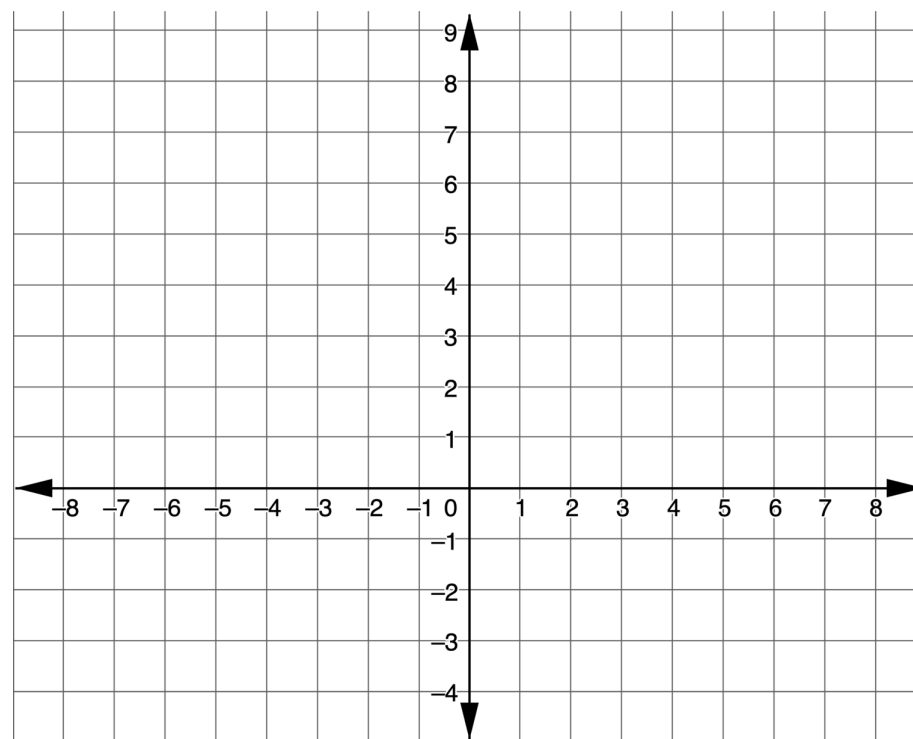
- a. $BHTX$ was created by
☐ reflection.
☐ translation.
☐ rotation.
☐ dilation.
 This transformation
☐ does
☐ does not
 preserve distance between the vertices.
- b. $KMPG$ was created by
☐ reflection.
☐ translation.
☐ rotation.
☐ dilation.
 This transformation
☐ does
☐ does not
 preserve distance between the vertices.
- c. $SGTW$ was created by
☐ reflection.
☐ translation.
☐ rotation.
☐ dilation.
 This transformation
☐ does
☐ does not
 preserve distance between the vertices.
- d. $WRCE$ was created by
☐ reflection.
☐ translation.
☐ rotation.
☐ dilation.
 This transformation
☐ does
☐ does not
 preserve distance between the vertices.



6.3.4 Guided Instruction: Sequence of Transformations and Distance Preservation

1. $GPMK$ with vertices $G(-6,3)$, $P(-4,5)$, $M(-3,3)$, and $K(-4,2)$ is reflected over $x = -2$, then translated right 2 units and up 1 unit to create $ZFVN$, respectively.

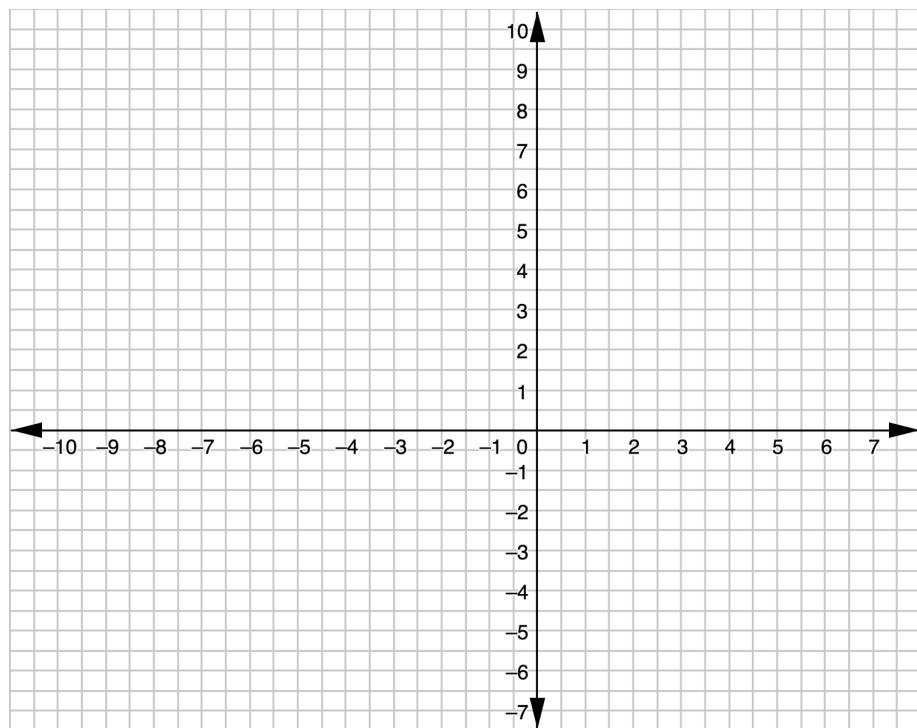
a. Draw the preimage $GPMK$ and the image $ZFVN$.



b. Use patty paper or the coordinates to determine if the corresponding sides of $GPMK$ and $ZFVN$ are congruent.

2. $THNK$ with vertices $T(-9, 3)$, $H(-7.5, 6)$, $N(-3, 3)$, and $K(-3, -6)$ is rotated 90° clockwise about the origin, then dilated using the scale factor of $\frac{1}{3}$ centered at the origin to create $T''H''N''K''$, respectively.

a. Draw the preimage $THNK$ and the image $T''H''N''K''$.



b. Determine if the corresponding sides of $THNK$ and $T''H''N''K''$ are congruent.

- c. Explain how you could use the description of a sequence of transformations to determine if the transformation does or does not preserve distance.

- d. How would your response differ if you were given the coordinates of a sequence of transformations?



6.3.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the lesson that I knew were ...	Something I wondered during the lesson was ...	Something I learned during the lesson was ...

Unit 6, Lesson 4: Sequence of Transformations



Benchmarks

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.



Learning Targets

- ☐ I can represent a sequence of transformations that will map the preimage onto the image.
- ☐ I can describe a sequence of transformations that will map the preimage onto the image.
- ☐ I can identify transformations that do or do not preserve distance.
- ☐ I can use a description of a sequence of transformations to determine if the transformation does or does not preserve distance.



6.4.1 Warm-Up: Would You Rather?

1. Would you rather drive a car at a rate of 40 miles per hour, or drive a car at a rate of 50 feet per second? Remember: There are 5280 feet in a mile.

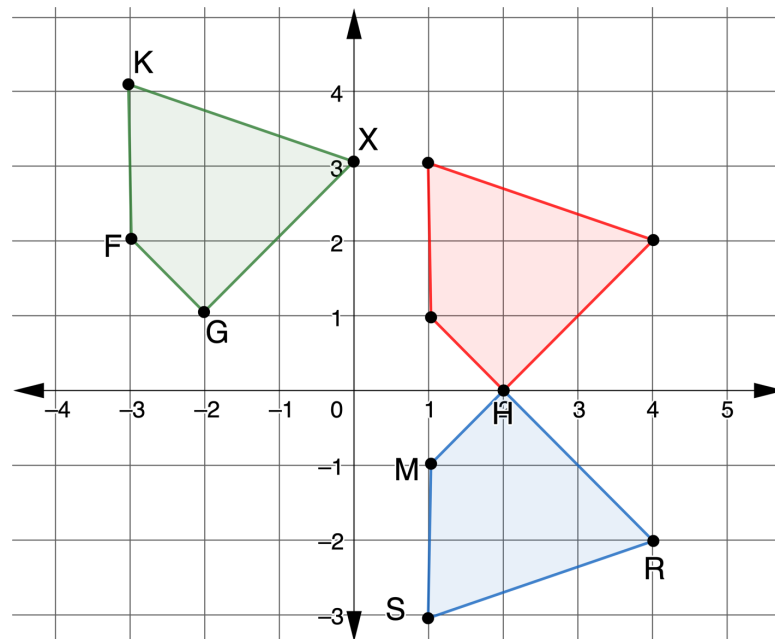


Explain your choice using mathematics.



6.4.2 Guided Instruction: Sequence of Transformations

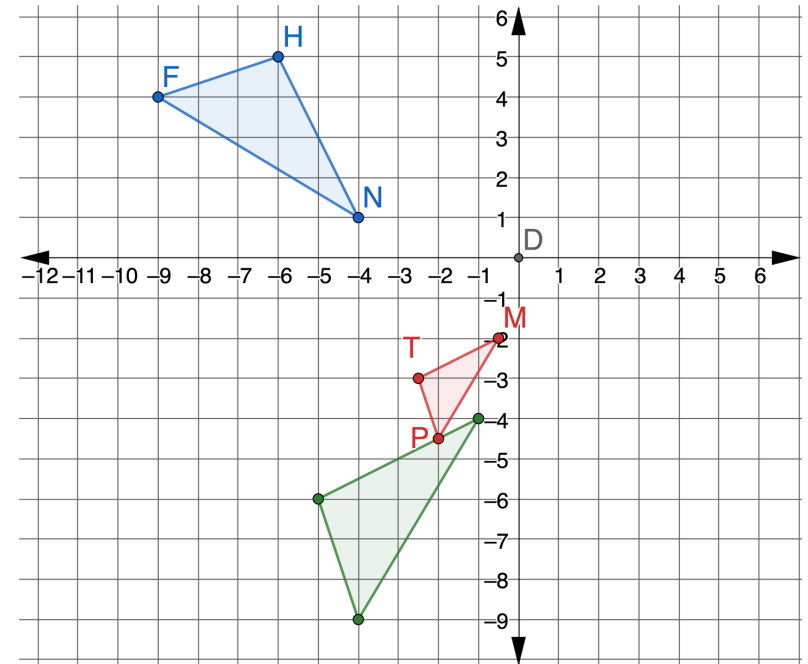
1. $XGFK$ is the result of a sequence of transformations of $RSMH$. The red quadrilateral is the first transformation.



- a. Describe the first transformation of $RSMH$.

- b. Write the transformation algebraically.
- c. Describe the second transformation of $RSMH$ that results in $XGFK$.
- d. Write the transformation algebraically.
- e. Explain if the sequence of transformations preserves distance.

2. $\triangle MTP$ is the result of a sequence of transformations of $\triangle HNF$.

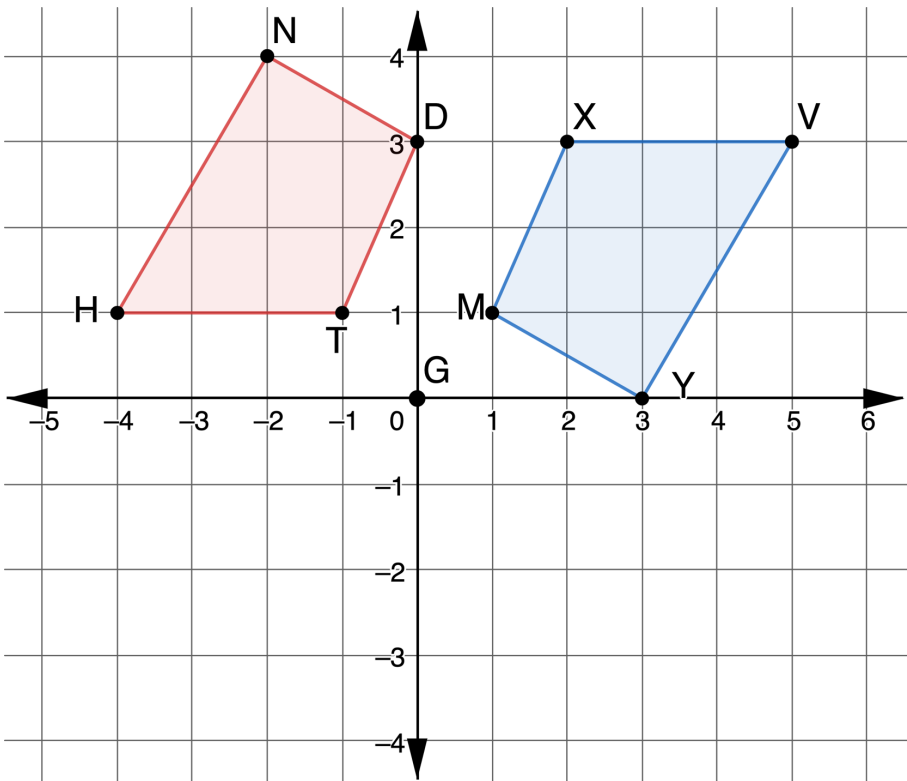


- a. Determine the sequence of transformations that will map $\triangle HNF$ onto $\triangle MTP$. Complete the table.

	$\triangle HNF$ Transformed to $\triangle H'N'F'$	$\triangle H'N'F'$ Transformed to $\triangle MTP$
Written Description		
Algebraic Description		

b. Explain how to determine if this sequence of transformations preserves distance.

3. $TDNH$ is the result of a sequence of transformations of XYV .



a. Determine the sequence of transformations that will map XYV onto $TDNH$. Complete the table.

	First Transformation	Second Transformation
Written Description		
Algebraic Description		

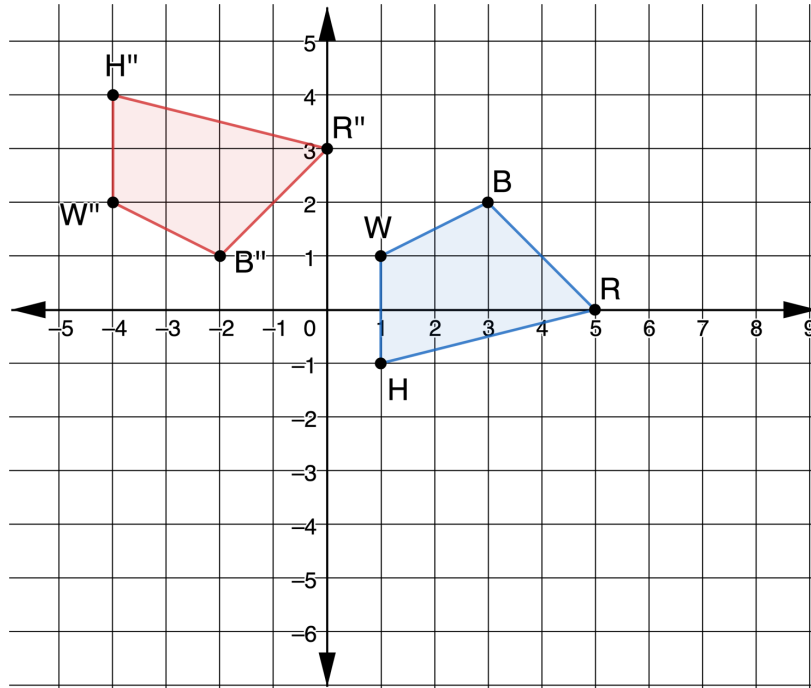
b. Which transformations preserve distance?





6.4.3 Guided Practice: Mapping a Given Preimage Onto an Image

1. $WBRH$ was transformed by a sequence of transformations to create $W''B''R''H''$.



Esmeralda and Wesley were determining the sequence of transformations and wrote their answers in the table.

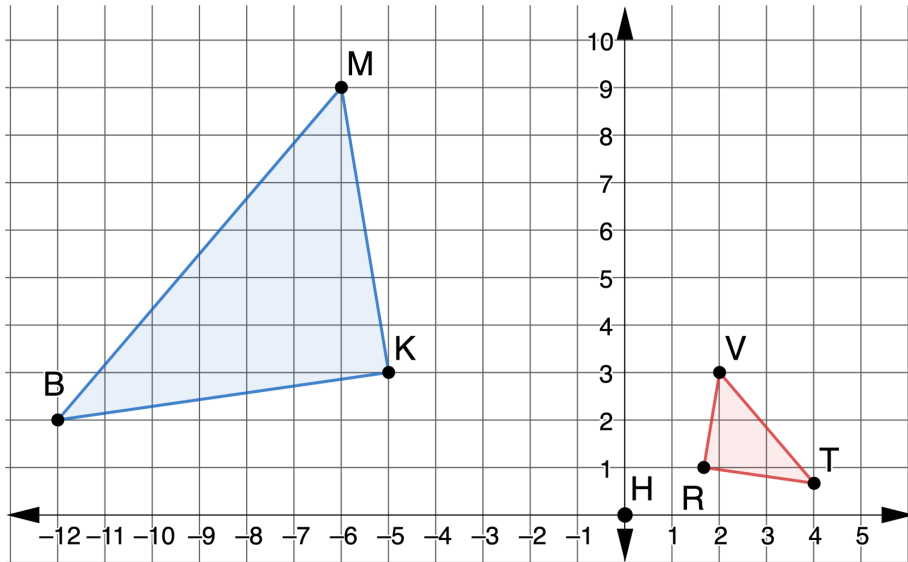
	$WBRH$ Transformed to $W'B'R'H'$	$W'B'R'H'$ Transformed to $W''B''R''H''$
Esmeralda's Answer	Translated 5 units to the left and 3 units down	Reflected over the x -axis
Wesley's Answer	Translated 3 units down	Rotated 180° about the origin

2. Determine whose sequence of transformations is correct. Explain your reasoning.



6.4.4 Your Turn!

1. $\triangle MKB$ was transformed to create $\triangle VRT$.

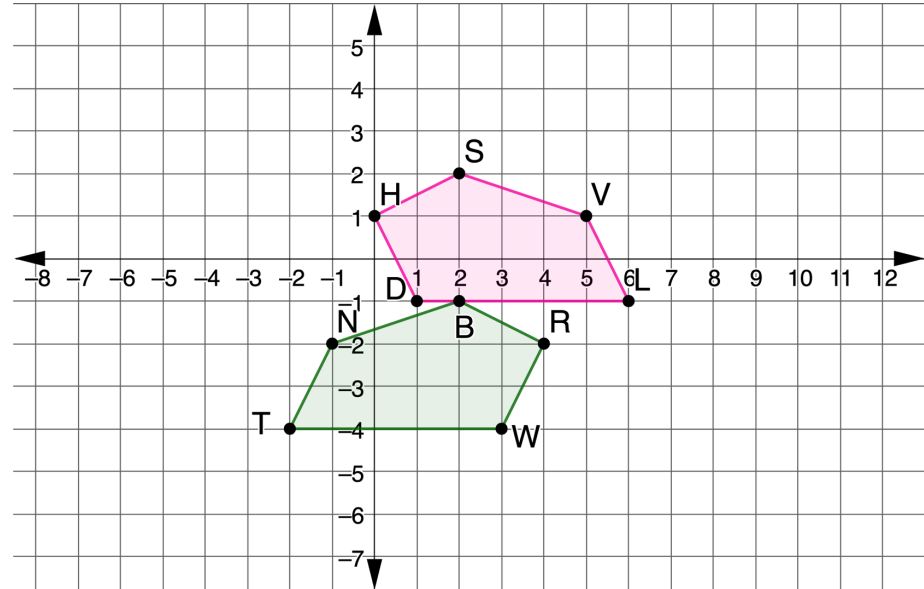


- Describe the sequence of transformations that will map $\triangle MKB$ onto $\triangle VRT$.
- Write the sequence of transformations using algebraic descriptions.



6.4.5 Wrap-Up: Ticket Out the Door

1. A sequence of transformations can map $NBRWT$ onto $VSHDL$.



Determine the sequence of transformations that will map $NBRWT$ onto $VSHDL$. Complete the table.

	First Transformation	Second Transformation
Written Description		
Algebraic Description		

Unit 6, Lesson 5: Drawing Transformations



Benchmark

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformation on a coordinate plane.



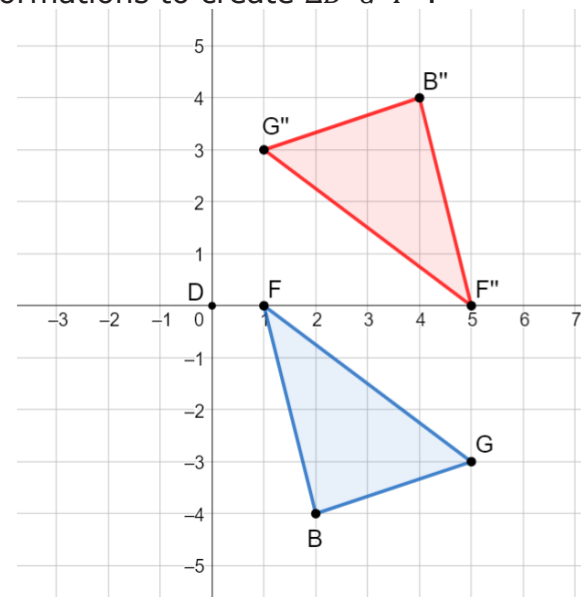
Learning Target

- ☐ I can draw the transformed figure when given a geometric figure and a sequence of transformations as a written description or an algebraic description.



6.5.1 Warm-Up: Error Analysis

- $\triangle BGF$ was transformed by a sequence of transformations to create $\triangle B''G''F''$.



Brian and Emily were determining the sequence of transformations and wrote their answers in the table.

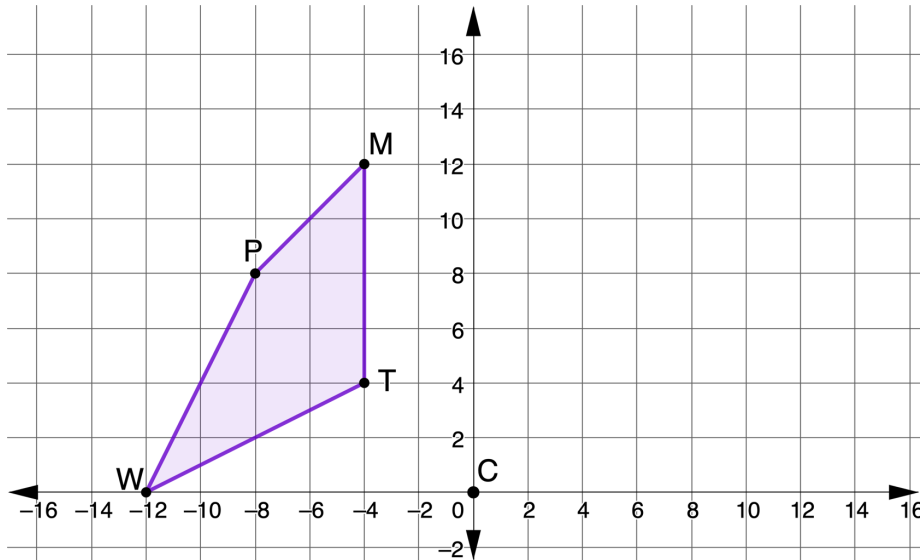
	$\triangle BGF$ Transformed to $\triangle B'G'F'$	$\triangle B'G'F'$ Transformed to $\triangle B''G''F''$
Brian's Answer	Rotated 180° about the origin	Translated right 6 units
Emily's Answer	Reflected over the x -axis	Translated right 4 units

Determine whose sequence of transformations is correct. Explain your reasoning.



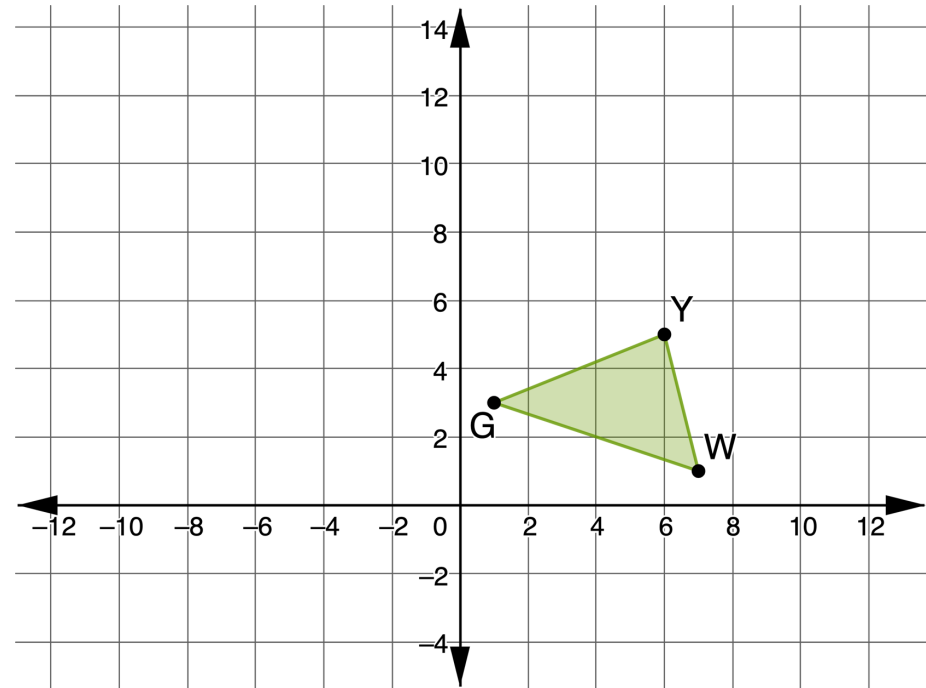
6.5.2 Guided Instruction: Drawing Transformations

1. $MTWP$ and point C are shown on the coordinate plane.



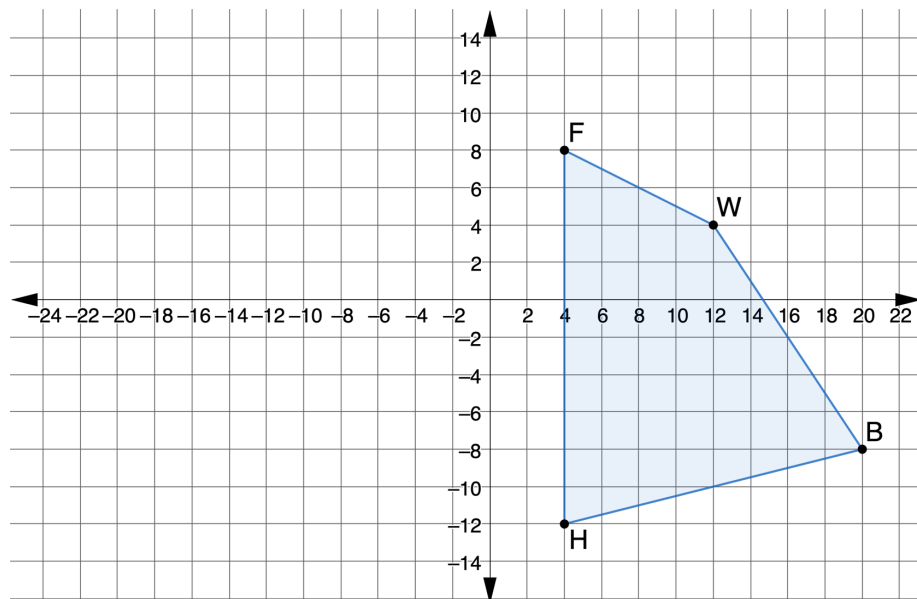
- Draw the quadrilateral that is a reflection of $MTWP$ over the y -axis on the coordinate plane above. Label the quadrilateral $M'T'W'P'$.
- Draw the quadrilateral that is a dilation of $M'T'W'P'$ centered at point C using a scale factor of $\frac{1}{4}$ on the coordinate plane above. Label the quadrilateral $M''T''W''P''$.
- Compare your drawings with your partner. Make any corrections if needed.

2. $\triangle GYW$ is shown on the coordinate plane.



- Draw $\triangle G'Y'W'$ as a result of $(x, y) \rightarrow (x - 4, y - 1)$ on the coordinate plane.
- Draw $\triangle G''Y''W''$ as a result of $(x, y) \rightarrow (3x, 3y)$ on the coordinate plane.

3. $FWBH$ is shown on the coordinate plane.

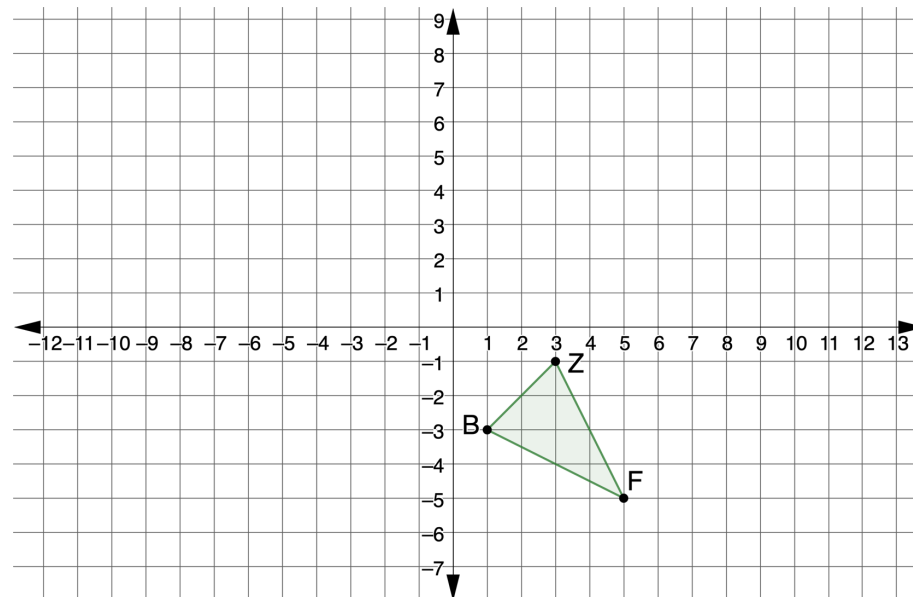


On the same coordinate plane, draw $F''W''B''H''$, which is a result of dilating $FWBH$ using a scale factor of $\frac{3}{4}$, with the center at point F , then rotating the quadrilateral 90° clockwise about point F .



6.5.3 Guided Practice: Sequence of Transformations

1. $\triangle FZB$ is shown on the coordinate plane.

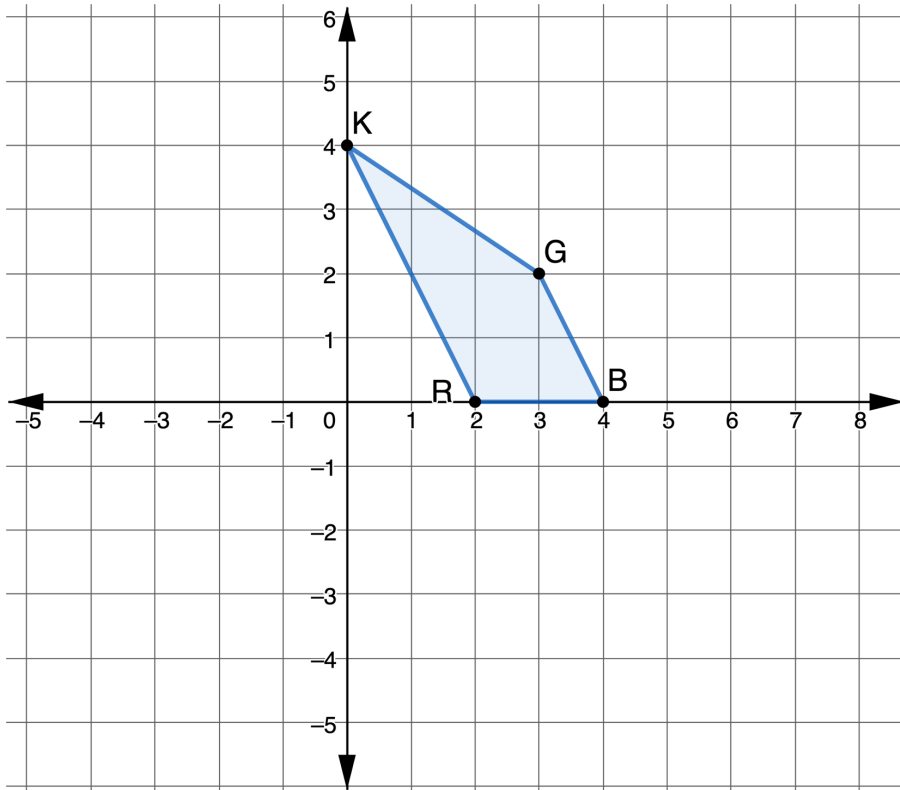


On the same coordinate plane, draw $\triangle F'''Z'''B'''$, which is the result of a reflection over the x -axis, then translated left 3 units and down 3 units, then dilated with a scale factor of $\frac{5}{2}$ using point Z'' as the center of dilation.



6.5.4 Your Turn!

1. $KGBR$ is shown on the coordinate plane.



On the same coordinate plane, draw $K''G''B''R''$, which is the result of $(x, y) \rightarrow (y, -x)$, then $(x, y) \rightarrow (\frac{1}{2}x, \frac{1}{2}y)$.



6.5.5 Wrap-Up: Cool Down

1. Why do you think it is important to use the prime notation when working with transformations?

Unit 6, Lesson 6: Similarity in Transformations



Benchmarks

MA.912.GR.2.8 Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar.

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.



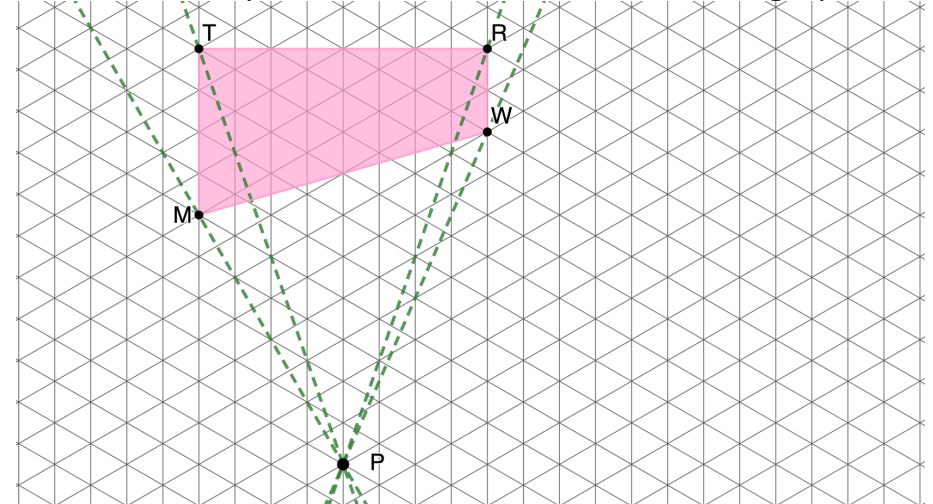
Learning Targets

- ☐ I can map one similar figure onto another and justify that the figures are similar.
- ☐ I can describe a sequence of transformations that will map the preimage onto the image.



6.6.1 Warm-Up: Dilation on an Isometric Graph

1. $TRWM$ and point P are drawn on an isometric graph.



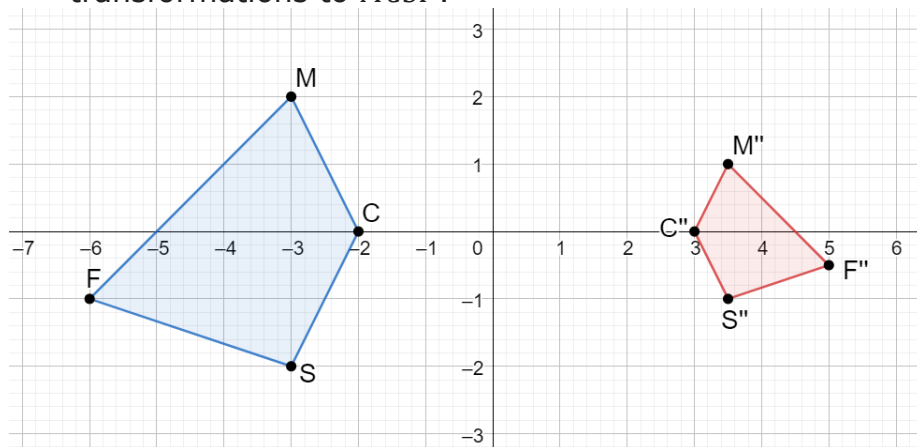
- a. Draw a dilation of $TRWM$ centered at point P using a scale factor of $\frac{1}{2}$.
- b. Explain what features on the graph were most useful in helping you determine where the dilation of $TRWM$ should be drawn.



6.6.2 Exploration: Similarity

With your partner, complete the following.

1. $M''C''S''F''$ was created by applying a sequence of transformations to $MCSF$.



- a. With patty paper, trace $\angle FMC$ in $MCSF$.
- b. Place the tracing over $\angle F''M''C''$ in $M''C''S''F''$.
- c. What do you notice about $\angle FMC$ and $\angle F''M''C''$?
- d. Repeat the tracing for $\angle C$, $\angle S$, and $\angle F$. What do you notice about each angle and their corresponding angles?

- e. Complete the table.

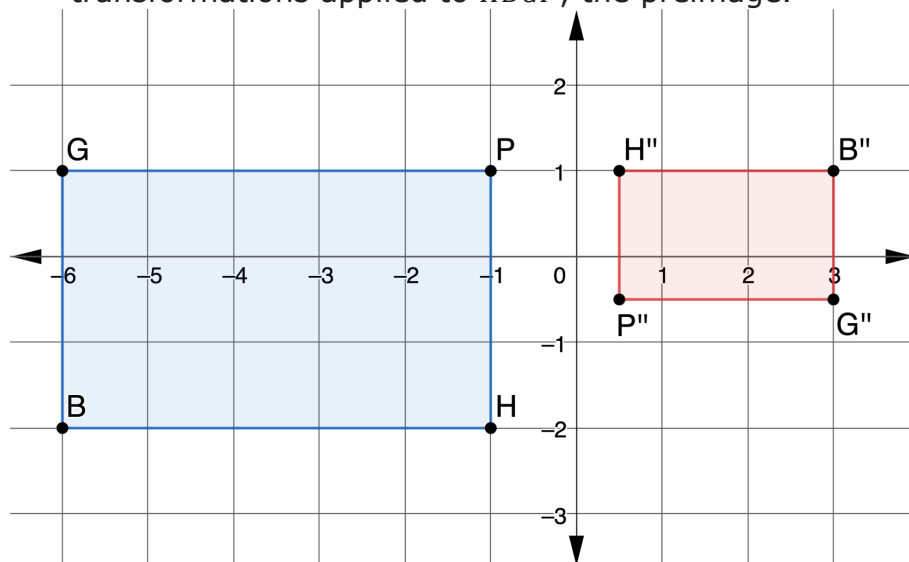
Angles	Comparison Between Angles
$\angle FMC$ and $\angle F''M''C''$	
$\angle MCS$ and $\angle M''C''S''$	
$\angle CSF$ and $\angle C''S''F''$	
$\angle SFM$ and $\angle S''F''M''$	

- f. Determine a sequence of transformations that can be used to map $MCSF$ to $M''C''S''F''$.
- g. Ask two classmates the transformations they determined. Record their transformations and write a summary of their reasoning. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating your summary was correct. Remind your classmates to include any details on the center of rotation or dilation and any scale factor.

Transformations	My Summary of Their Reason	Initials

- h. Work with your partner to verify your classmates' transformations are correct.

2. $H''B''G''P''$ was created using a sequence of transformations applied to $HBGP$, the preimage.



- a. Complete the table.

Preimage	Side Length	Image	Side Length	Ratio of Preimage to Image Lengths
GP		$G''P''$		
PH		$P''H''$		
HB		$H''B''$		
BG		$B''G''$		

- b. Describe the relationship between the ratios of the preimage to the image.

- c. Determine the sequence of transformations that was used to create $H''B''G''P''$ from $HBGP$.

- d. Discuss with your partner how to describe how $MCSF$ is related to $M''C''S''F''$ and how $HBGP$ is related to $H''B''G''P''$. Summarize your discussion.

3. What qualities of similar figures do these quadrilaterals have?



Similarity is having exactly the same shape but not necessarily the same size. Two figures are similar if and only if one figure can be obtained from the other by a dilation or a sequence of transformations that includes a dilation.

4. Complete the statements using words from the bank.

Word Bank			
congruent	dilation	proportional	similar

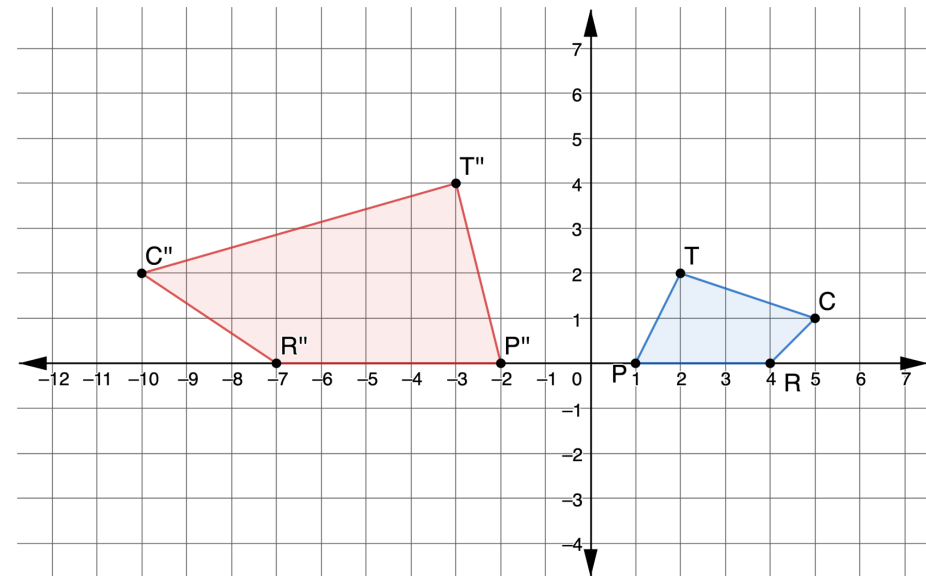
When a sequence of transformations includes a _____, the resulting image of the sequence of transformations is _____ to the preimage. All of the angles of the resulting image are _____ to the corresponding angles of the preimage. The corresponding sides of the image to the preimage are _____.



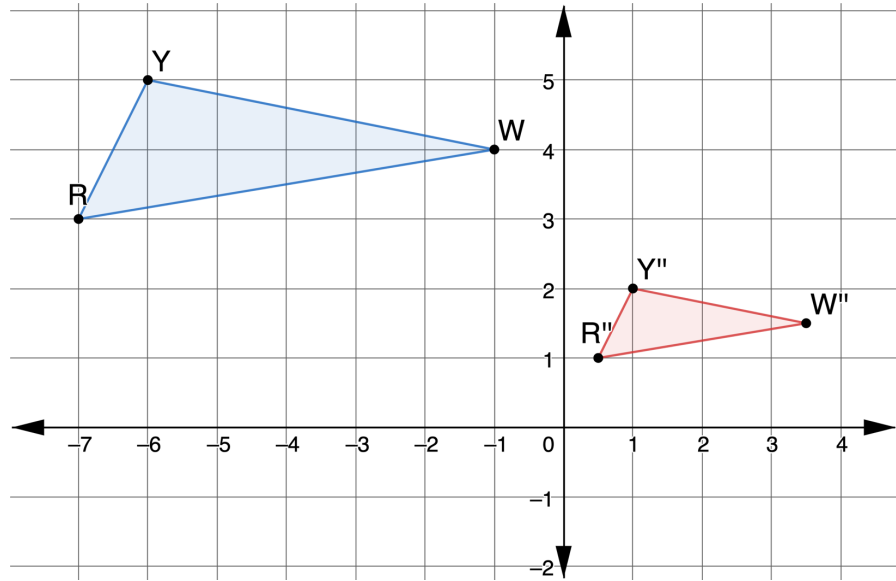
6.6.3 Guided Practice: Similarity

For each pair of figures, write the sequence of transformations that would show the figures are similar. If the figures are not similar, explain why.

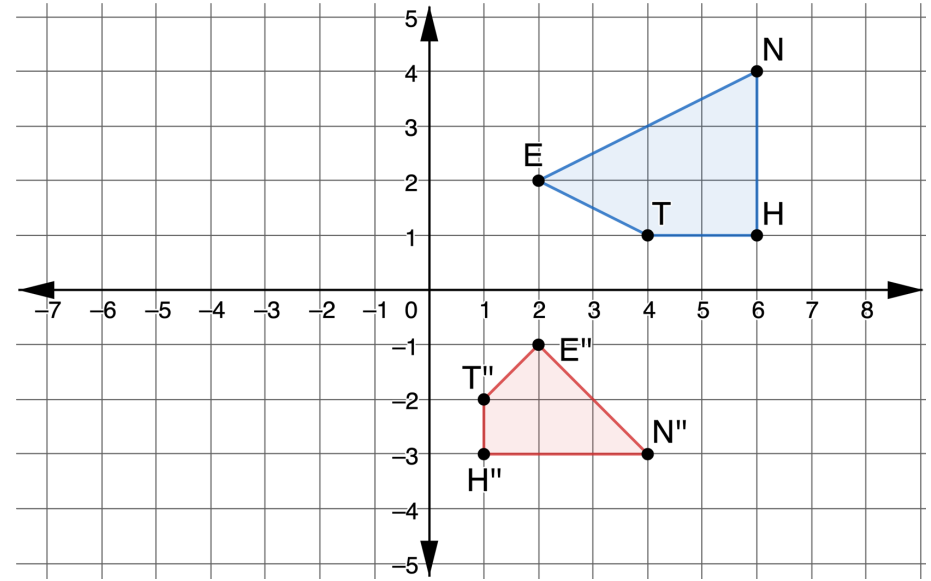
1.



2.



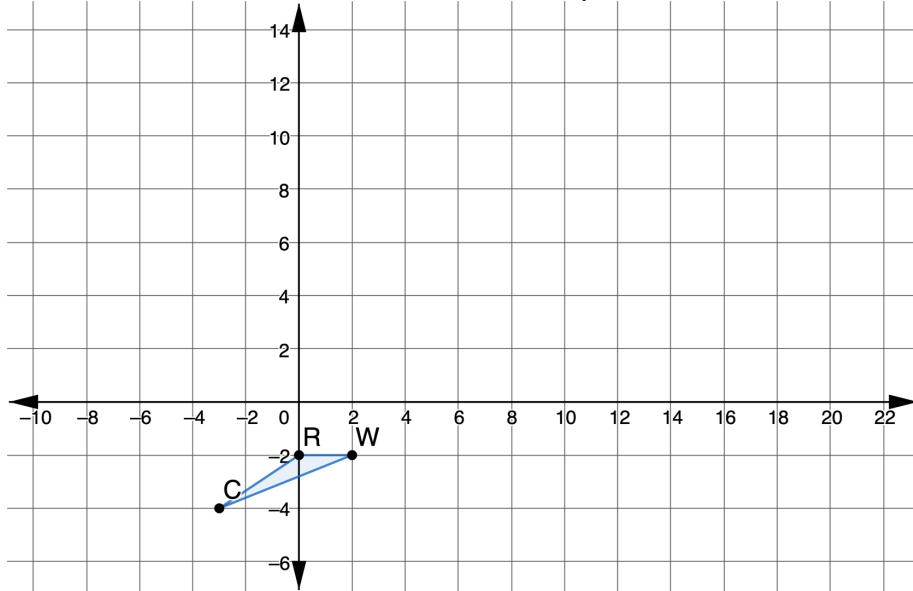
3.





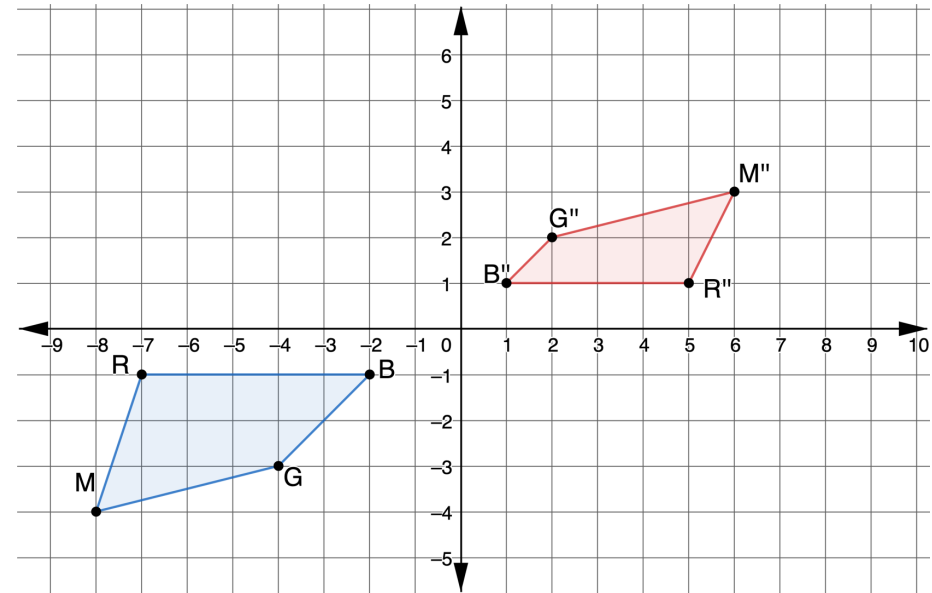
6.6.4 Your Turn!

1. $\triangle RWC$ is shown on the coordinate plane.



- Draw $\triangle R''W''C''$, which is a result of a translation up 4 units and left 1 unit, then dilated using a scale factor of 5 centered at C' .
- Explain how $\triangle RWC$ is similar to $\triangle R''W''C''$.

2. Carter and Isabella are asked to determine if the quadrilaterals are similar. Their responses are recorded in the table.



Carter's Response	Isabella's Response
Yes, the quadrilaterals are similar because all the corresponding angles are congruent.	No, the quadrilaterals are not similar because the corresponding sides are not proportional.

Is Carter's or Isabella's response correct? Explain your decision.



6.6.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to help them understand how to determine if a sequence of transformations results in similar figures.

Unit 6, Lesson H1: Proving All Circles Are Similar



Benchmark

MA.912.GR.6.5 Apply transformations to prove that all circles are similar.

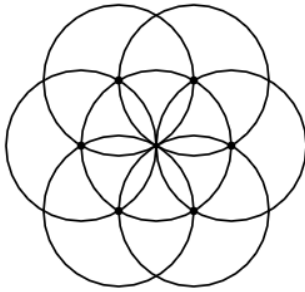


Learning Target

- ☐ I can justify that two circles are similar and conclude that any circle can be mapped to another circle using similarity transformations.



6.H1.1 Warm-Up: Notice and Wonder



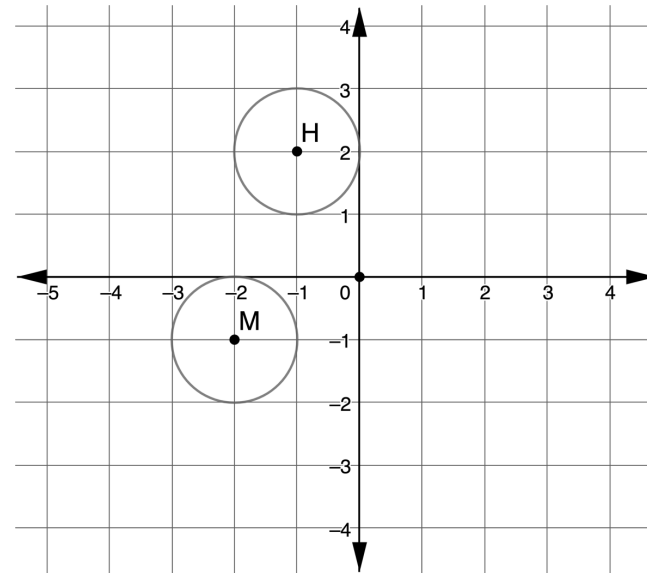
1. What do you notice?
2. What do you wonder?



6.H1.2 Exploration: Are Circles Similar?

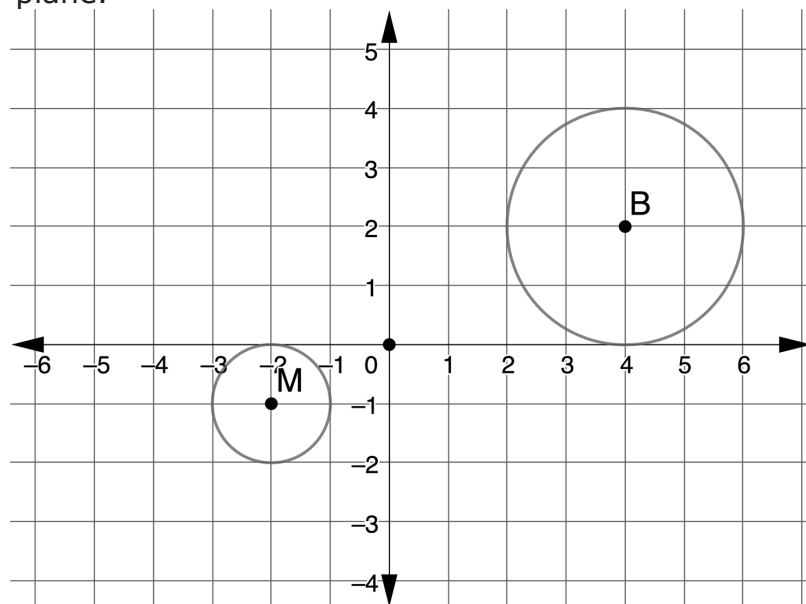
Answer the following questions with your partner.

1. Circle M and Circle H are shown on the coordinate plane.



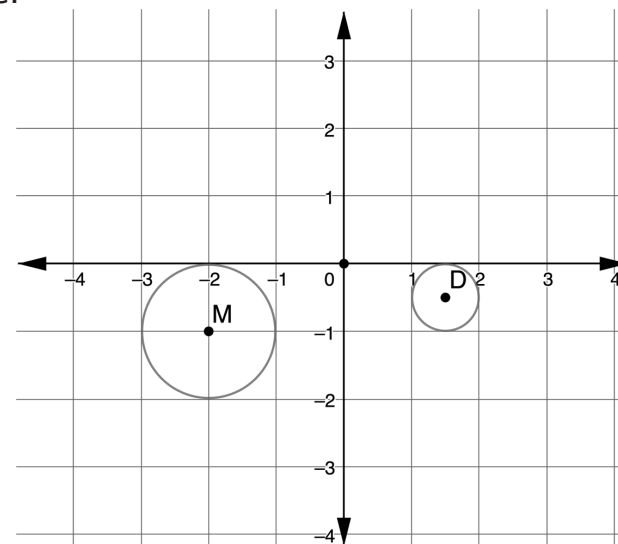
- a. How can Circle M be mapped onto Circle H ?
- b. What can you conclude about Circle M and Circle H ?

2. Circle M and Circle B are shown on the coordinate plane.



- How can Circle M be mapped onto Circle B ?
- What can you conclude about Circle M and Circle B ?

3. Circle M and Circle D are shown on the coordinate plane.

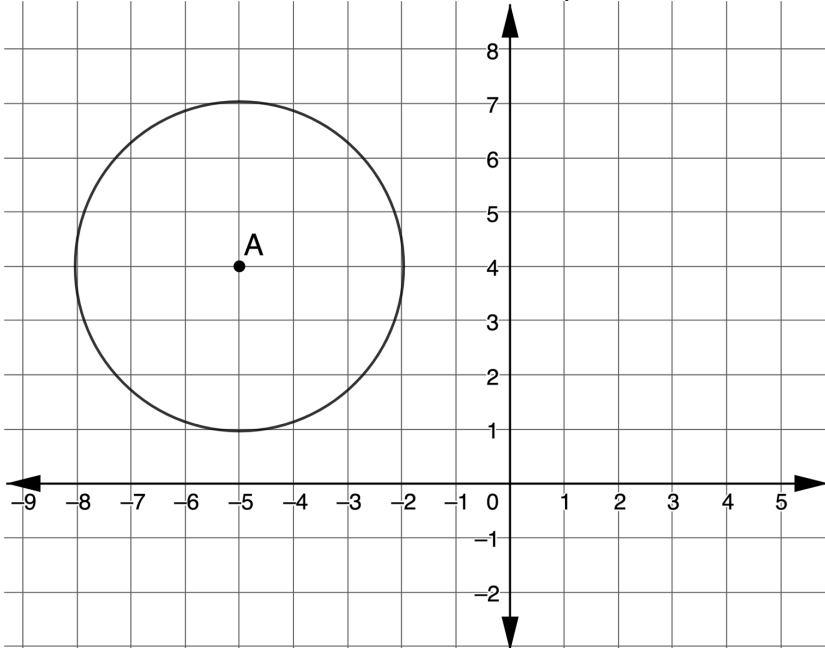


- How can Circle M be mapped onto Circle D ?
 - What can you conclude about Circle M and Circle D ?
4. What can you conclude about Circle M , Circle H , Circle B , and Circle D ?
5. Complete the statement.
- | | | | |
|---|-------------|--|-------------|
| ☐ All
☐ Some
☐ None | circles are | ☐ congruent
☐ similar | because any |
|---|-------------|--|-------------|
- circle can be mapped to another circle using similarity transformations.



6.H1.3 Your Turn!

1. Circle A is shown on the coordinate plane.



- a. Draw Circle P , the result of a translation 5 units right and 2 units down and a dilation of $\frac{1}{2}$ about the center of circle A .
- b. Explain how Circle A and Circle P are similar. Include the word "transformation" in your explanation.

2. Lana and Henry are writing notes from today's lesson.

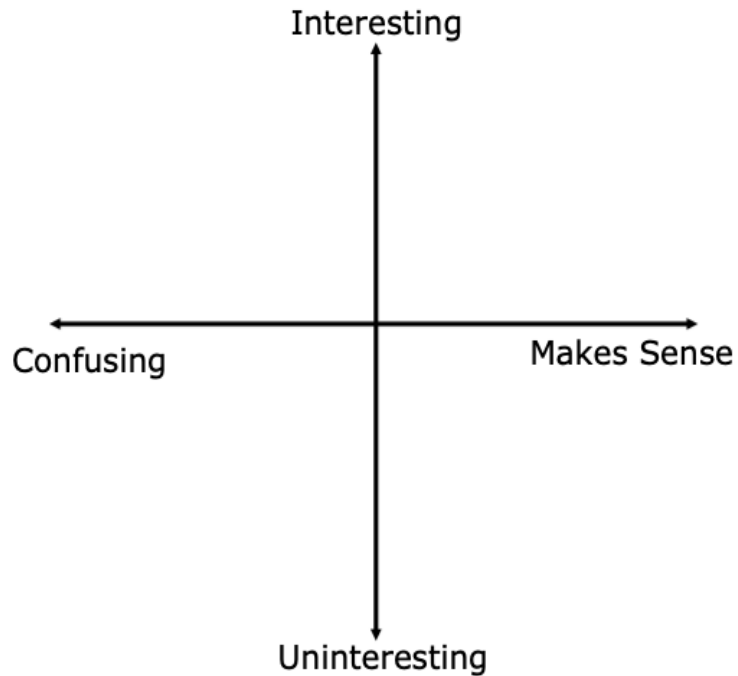
Lana's Notes	Henry's Notes
All circles are similar; therefore, all circles are also congruent.	All circles are similar, but not all circles are congruent.

Whose notes do you agree with? Why?



6.H1.4 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest and understanding for today's learning target.



Explain the location of your point.

Unit 6, Lesson H2: Proofs by Contradiction



Benchmark

MA.912.LT.4.8 Construct proofs, including proofs by contradiction.



Learning Target

- ☐ I can use proof by contradiction to justify that all circles are similar.



6.H2.1 Warm-Up: Critical Thinking

1. DirecTV commercials are entertaining because the conditional statements are outrageous. Write five conditional statements with an outrageous conclusion that could be used for a DirecTV commercial.



6.H2.2 Exploration: Contradictions

In the DirecTV commercial, there were seven conditional statements before the final conclusion.

Order	Statement
1	If your cable company puts you on hold, then you get angry.
2	If you get angry, then you go blow off steam.
3	If you go blow off steam, then an accident happens.
4	If an accident happens, then you get an eye patch from the doctor.
5	If you get an eye patch, then people think you are tough.
6	If people think you are tough, then they want to fight you.
7	If people want to fight you, then you get hurt.
8	You will end up in a roadside ditch.

1. Assume that the conclusion is not true. Write a contradiction to the conclusion.
2. Share your contradiction with your partner.

3. Use your contradiction to write a “rewind” for the commercial with statements that would lead to the contradiction.

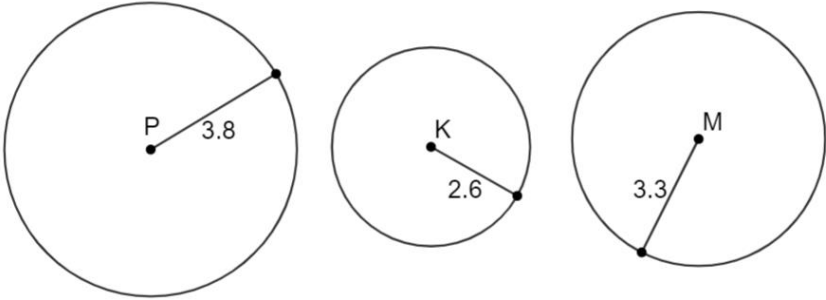
Order	Statement
8	
7	
6	
5	
4	
3	
2	
1	

4. Explain how writing the contradiction would change the commercial.



6.H2.3 Exploration: Circles Are Similar

1. Circles P , K , and M are shown. The radius of Circle P is 3.8 units long. The radius of Circle K is 2.6 units long. The radius of Circle M is 3.3 units long.



- a. With your partner, fill in the table. Round to nearest thousandth. Use the π button on your calculator.

Circle	Radius	Circumference

- b. Complete the table.

Circle P and K	
Ratio of Radius of Circle P and Radius of Circle K	
Ratio of Circumference of Circle P and Circumference of Circle K	

c. What do you notice about the ratios?

d. Complete the table.

Circle K and M	
Ratio of Radius of Circle K and Radius of Circle M	
Ratio of Circumference of Circle K and Circumference of Circle M	

e. What do you notice about the ratios?

2. Make a conjecture about the ratios of radii and circumferences of circles.



6.H2.4 Guided Instruction: Proof by Contradiction

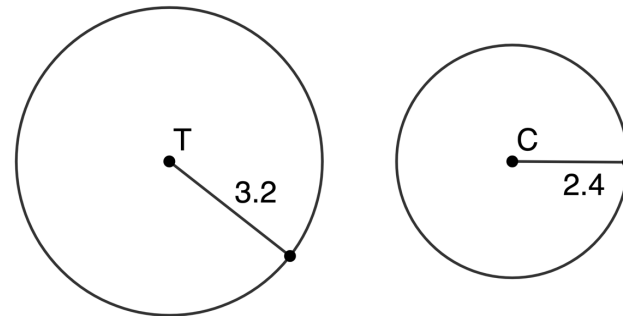
A way to write a proof is by using a contradiction.



Proof by Contradiction

Assume the statement to be proven is false, and work to show its falsity until the result of that assumption is a contradiction.

1. Circles T and C are shown. The radius of Circle T is 3.2 units long. The radius of Circle C is 2.4 units long.



Prove that Circle T is similar to Circle C .

a. Write an assumption that the statement is false.

b. Find the ratio of the radius of Circle T to the radius of Circle C .

- c. Find the ratio of the circumference of Circle T to the circumference of Circle C .

- d. Explain how the ratios show the assumption is not true.

- e. Complete the proof.

Assume that Circle T is _____ to Circle C . The ratio of the radius of Circle T to the radius of Circle C is _____. The ratio of the circumference of Circle T to the circumference of Circle C is _____. The ratio of the radii and the ratio of the circumferences are _____, so corresponding parts of Circle T and Circle C are proportional. Since the corresponding parts of Circle T and Circle C are proportional, the circles are _____. This is a contradiction, so the assumption that Circle T is not similar to Circle C is not true.



6.H2.5 Wrap-Up: Reflection

1. Check below all the statements that apply to your understanding.
 - ☐ I understand how to write a contradiction.
 - ☐ I need support on how to write a contradiction.
 - ☐ I understand that all circles are similar.
 - ☐ I need support on understanding why all circles are similar.
 - ☐ I understand how to prove that all circles are similar by contradiction.
 - ☐ I need support on how to prove that all circles are similar by contradiction.

Unit 6, Lesson 7: Triangle Similarity



Benchmark

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.



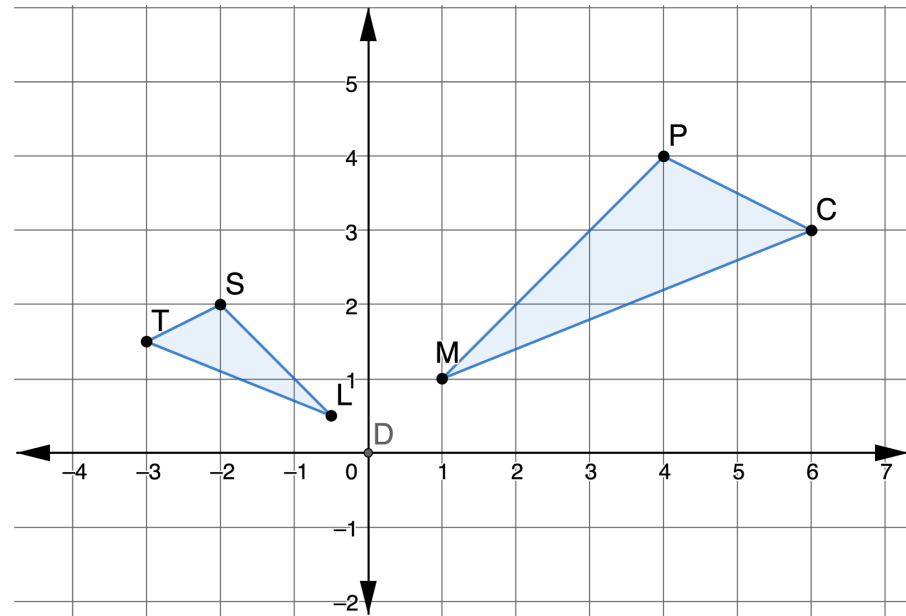
Learning Targets

- ☐ I can identify similar triangles and write similarity statements.
- ☐ I can use given information to show that two triangles are similar.



6.7.1 Warm-Up: Bell Work

- $\triangle MPC$ and $\triangle TSL$ are shown on the coordinate grid, where $\triangle MPC$ is the pre-image.



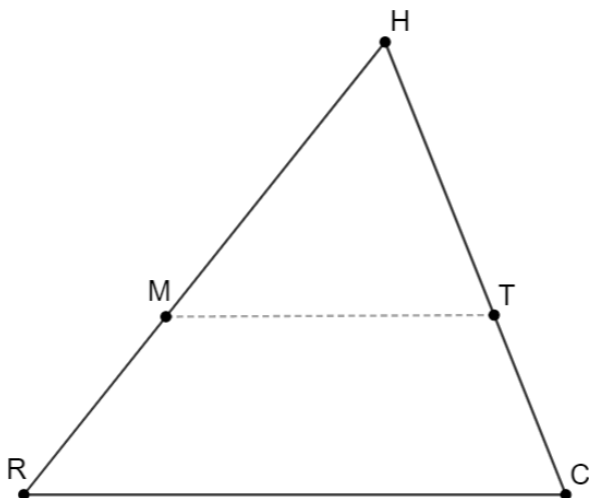
Describe a sequence of two transformations that will map the triangles.



6.7.2 Exploration: Similar Triangles

With your partner, complete the following.

$\triangle HCR$ is shown, where $m\angle HCR = 68^\circ$, $m\angle HRC = 51^\circ$, and $m\angle RHC = 61^\circ$. Point M is on $\overline{HR}$ and point T is on $\overline{HC}$. $\overline{MT}$ is drawn as a dotted line.



1. Use a ruler and a protractor to complete the table.

Measurements of $\triangle RHC$		Measurements of $\triangle MHT$	
$RH =$	$m\angle HRC = 51^\circ$	$MH =$	$m\angle HMT =$
$HC =$	$m\angle RHC = 61^\circ$	$HT =$	$m\angle MHT = 61^\circ$
$RC =$	$m\angle HCR = 68^\circ$	$MT =$	$m\angle HTM =$

2. What do you notice about the corresponding angles of $\triangle RHC$ and $\triangle MHT$?

3. Discuss with your partner what relationship could have been given so that it would not have been necessary to use a protractor to find $m\angle HMT$ and $m\angle HTM$. Summarize your discussion.

4. What do you notice about the corresponding sides of $\triangle RHC$ and $\triangle MHT$?

5. Discuss with your partner how $\triangle RHC$ and $\triangle MHT$ are related. Summarize your findings.

6. Complete the statement that shows the relationship. $\triangle RHC$ is similar to $\triangle$ _____.

7. Complete each relationship for $\triangle RHC$ and $\triangle MHT$.

$$\frac{RH}{TM} = \frac{RC}{MT}$$

$$\frac{HT}{CR} = \frac{HM}{MT}$$

8. Discuss with your partner a sequence of transformations that can be used to map $\triangle RHC$ onto $\triangle MHT$. Summarize your discussion by describing the sequence.

9. Compare your sequence of transformations with another pair of students. Ask two classmates for their sequence of transformations. Record their sequence and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating your summary is correct.

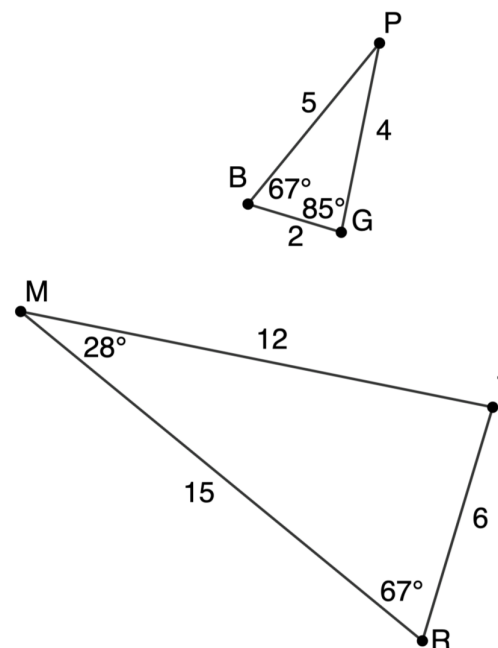
Sequence	My Summary of Their Reason	Initials

10. What transformation ensures that the sequence of transformations will result in similar triangles?



6.7.3 Your Turn!

1. $\triangle PGB$ and $\triangle MTR$ are shown. Use the given information to determine if $\triangle PGB$ is similar to $\triangle MTR$. If they are similar, write a similarity statement. If they are not, explain why.





6.7.4 Activity: Stations

1. With your partner, create two triangles using the description provided. Try to make the two triangles as different as possible.

Description Letter	Drawing

Description Letter	Drawing

Description Letter	Drawing

Description Letter	Drawing

2. Which descriptions resulted in similar triangles?





Angle-Angle Similarity (AA~) Theorem

If two angles of one triangle are congruent to two angles of another triangle, then the two triangles are similar.



Side-Side-Side Similarity (SSS~) Theorem

If the lengths of the corresponding sides of two triangles are proportional, then the triangles are similar.



Side-Angle-Side Similarity (SAS~) Theorem

If the lengths of two sides are proportional and their included angles are congruent on two different triangles, then the triangles are similar.

3. Match each description with the postulate or theorem that describes it. If none describe it, then write "none."

Description Letter	Theorem



6.7.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Unit 6, Lesson 8: Proving Triangle Similarity – Part 1



Benchmark

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.



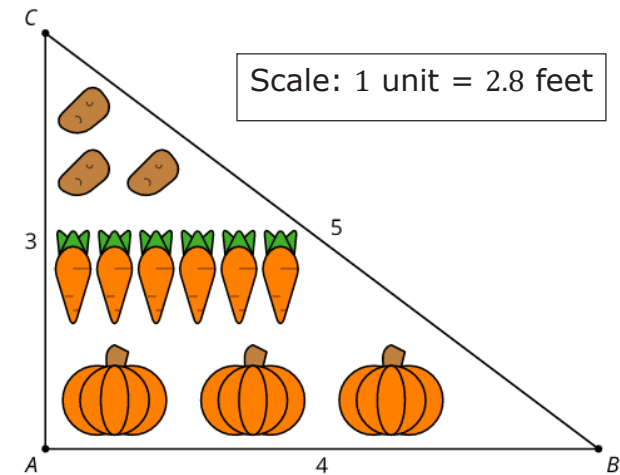
Learning Target

- ☐ I can use triangle similarity to prove triangles are similar.



6.8.1 Warm-Up: Vegetable Garden

- $\triangle ABC$ is used to create a plan for a vegetable garden.



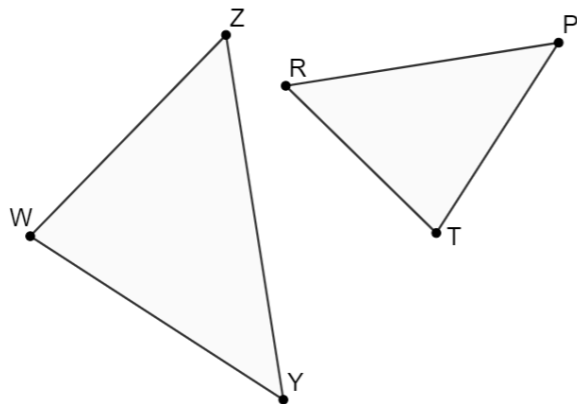
- What do you notice about the plan?
- Write at least three equivalent ratios or equations using lengths from both the diagram and the full-size garden.



6.8.2 Collaboration: Proving Triangle Similarity

With your partner, answer the following questions.

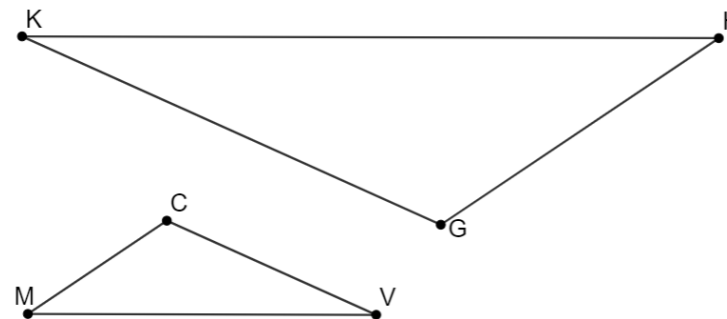
1. $\triangle WZY$ and $\triangle RPT$ are shown. $\angle WZY \cong \angle TRP$ and $\angle ZWY \cong \angle RTP$.



- Mark the figure using the given information.
- What similarity theorem can be applied to show that the triangles are similar?
- Write the similarity statement.
- Complete each.

$$\frac{WZ}{PT} = \frac{ZY}{RT} \quad \angle WYZ \cong$$

2. $\triangle KHG$ and $\triangle MVC$ are shown. $\angle KHG \cong \angle VMC$ and $\frac{KH}{VM} = \frac{GH}{CM}$.



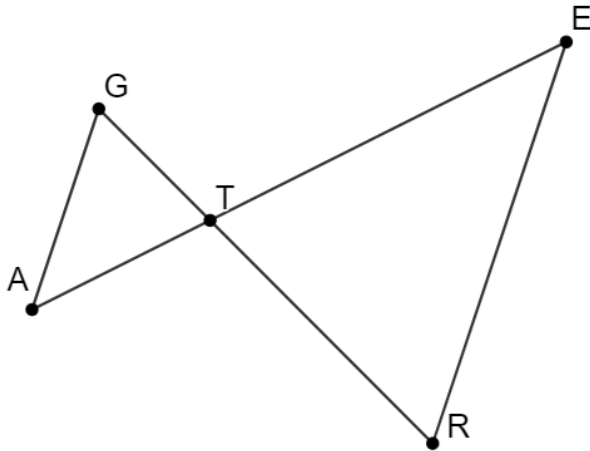
- Mark the figure using the given information.
- What similarity theorem can be applied to show that the triangles are similar?
- Write the similarity statement.
- Complete each.

$$\frac{KG}{CM} = \frac{GH}{CM} \quad \angle HGK \cong \quad \angle CVM \cong$$



6.8.3 Guided Instruction: Proving Triangle Similarity

1. $\triangle GTA$ and $\triangle TER$ are shown, and $\overline{GA} \parallel \overline{ER}$.



- a. With your partner, discuss what conclusions can be drawn using the given information. Summarize your discussion and explain how you reached each conclusion.
- b. Which similarity postulates should be used to prove $\triangle GTA \sim \triangle RTE$?

- c. Complete the two-column proof.

Statement	Reason
1. $\overline{GA} \parallel \overline{ER}$	1.
2.	2. Alternate Interior Angles are congruent
3.	3.
4. $\triangle GTA \sim \triangle RTE$	4.

- d. Ask two classmates about their proof. Record their step 3 statement and reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Step 3 Statement	Step 3 Reason	Initials

- e. Discuss with your partner the two different ways that $\triangle GTA \sim \triangle RTE$ can be proved.

- f. Explain why you do not need to state three congruent angles in your two-column proof.

- g. Arian and Isabella were discussing why there is no Angle-Angle-Angle Similarity Theorem.

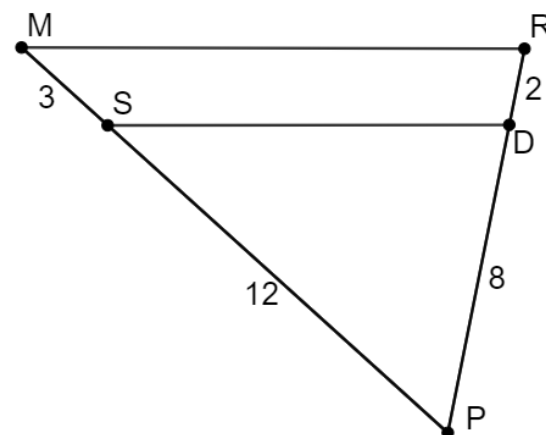
Arian	Isabella
I think once you know two pairs of angles in a triangle are congruent, then you know the last pair are congruent since the sum of all of the angles of a triangle are 180° .	I think once you know two pairs of angles in a triangle are congruent, then you know all of the sides are proportional.

Discuss with your partner why Arian and Isabella's arguments could also be applied to show why Angle-Side-Angle similarity and Angle-Angle-Side similarity is not needed to prove similarity. Summarize your discussion.

2. $\triangle SPD$ and $\triangle MPR$ are shown.

Given: $MS = 3$, $SP = 12$, $RD = 2$, and $DP = 8$

Prove: $\triangle SDP \sim \triangle MRP$

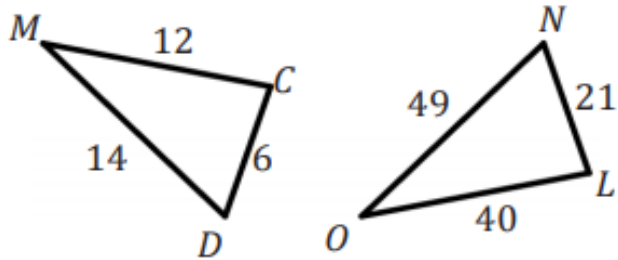


Statement	Reason
1. $MS = 3$, $SP = 12$, $RD = 2$, and $DP = 8$	1.
2. $MS + SP = PM$ $RD + DP = RP$	2.
3. $PM = 15$ $RP = 10$	3. Substitution
4. $\frac{PS}{PM} = \frac{12}{15}$ $\frac{PD}{PR} = \frac{8}{10}$	4.
5. $\frac{PS}{PM} = \frac{PD}{PR}$	5.
6. $\angle P \cong \angle P$	6.
7. $\triangle SDP \sim \triangle MRP$	7.



6.8.4 Try It: Triangle Similarity

1. Amanda and Tyler are determining if $\triangle MCD \sim \triangle OLN$ is true. Their reasonings are shown.

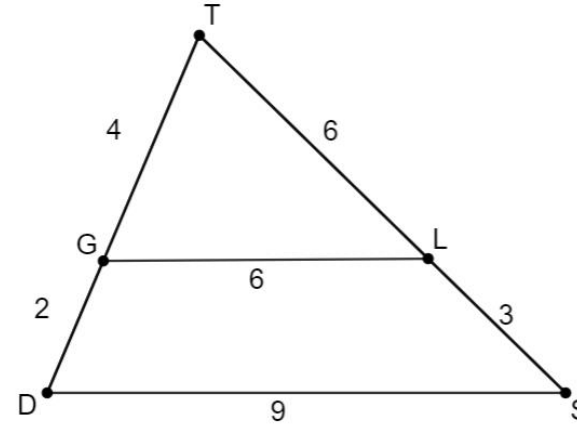


Amanda's Answer	Tyler's Answer
No, the triangles are not similar because the corresponding sides are not proportional.	Yes, the triangles are similar because the corresponding sides are proportional.

Who do you agree with? Explain your reasoning.

2. $\triangle TLG$ and $\triangle TSD$ are shown.

Given: $TG = 4$, $GD = 2$, $TL = 6$, $LS = 3$, $GL = 6$, and $DS = 9$
Prove: $\triangle TLG \sim \triangle TSD$

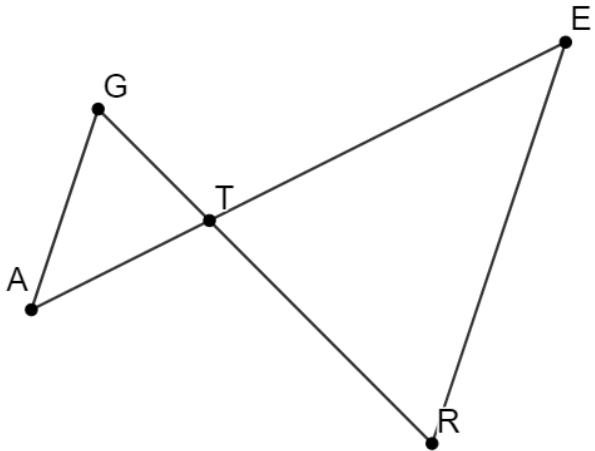


Statement	Reason
1. $TG = 4$, $GD = 2$, $TL = 6$, $LS = 3$, $GL = 6$, and $DS = 9$	1.
2. $TG + GD = TD$ $TL + LS = TS$	2.
3. $TD = 6$, $TS = 9$	3. Substitution
4. $\frac{TG}{TD} = \frac{4}{6}$, $\frac{TL}{TS} = \frac{6}{9}$, $\frac{GL}{DS} = \frac{6}{9}$	4.
5. $\frac{TG}{TD} = \frac{TL}{TS} = \frac{GL}{DS}$	5.
6. $\triangle TLG \sim \triangle TSD$	6.



6.8.5 Wrap-Up: Ticket Out the Door

1. $\triangle GTA$ and $\triangle TER$ are shown.



Given: $TA = \frac{2}{3}TE$ and $TG = \frac{2}{3}RT$

Prove: $\triangle GTA \sim \triangle RTE$

Complete the two-column proof.

Statement	Reason
1. $TA = \frac{2}{3}TE$ and $TG = \frac{2}{3}RT$	1. Given
2. $\frac{TA}{TE} = \frac{2}{3}$, $\frac{TG}{RT} = \frac{2}{3}$	2.
3.	3. Transitive Property
4. $\angle GTA \cong \angle RTE$	4.
5. $\triangle GTA \sim \triangle RTE$	5.



Unit 6, Lesson 9: Proving Triangle Similarity – Part 2



Benchmark

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse Leg.



Learning Target

- ☐ I can use triangle similarity to prove that triangles are similar.



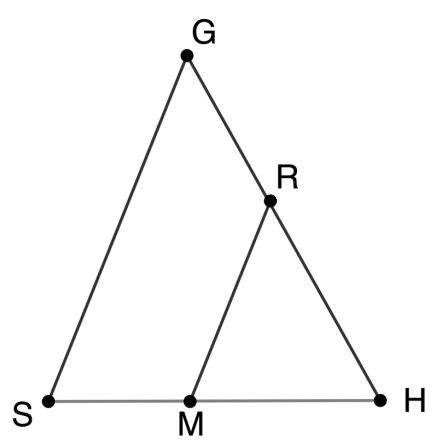
6.9.1 Warm-Up: Would You Rather?

1. In 2021, General Motors introduced the eVTOL air-taxi and the PAV (personal autonomous vehicle), a self-driving car. Both are concepts with no known timeline for production.
 - a. Do you think a drone-like personal vehicle like the eVTOL air-taxi will be reality in your lifetime?
 - b. Which would you rather have as your vehicle, an air-taxi or a self-driving car, in the future?
 - c. Explain your choice.



6.9.2 Activity: Proving Triangle Similarity

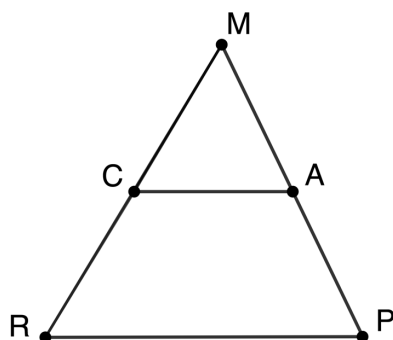
With your partner, sort through the statement and reason cards. Then, write each proof in the tables provided.

Proof A	
 Given: $\overline{GS} \parallel \overline{RM}$ Prove: $\triangle RHM \sim \triangle GHS$	
Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.

Proof B

Given: $MC = \frac{2}{5}MR$ and
 $MA = \frac{2}{5}MP$

Prove: $\triangle MCA \sim \triangle MRP$

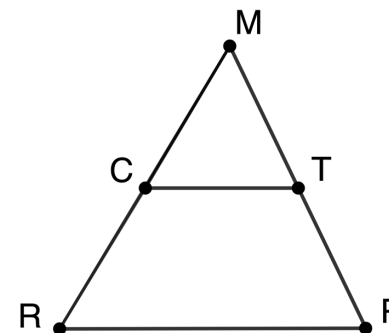


Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.

Proof C

Given: $\angle RPT$ and $\angle CTP$ are supplementary

Prove: $\triangle MCT \sim \triangle MRP$

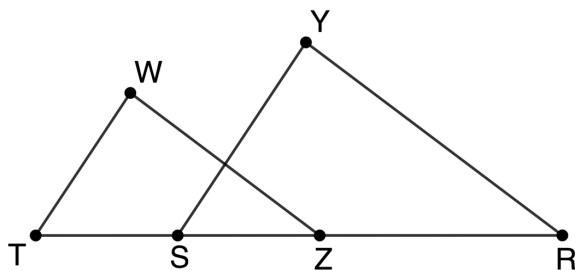


Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7.	7.
8.	8.
9.	9.

Proof D

Given: $\overline{WT} \parallel \overline{YS}$,
 $\overline{WZ} \parallel \overline{YR}$

Prove:
 $\triangle WZT \sim \triangle YRS$

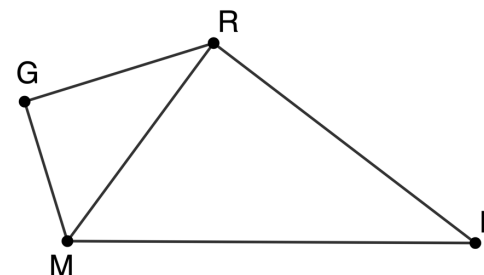


Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.

Proof E

Given: $GM = \frac{3}{5}RM$,
 $RG = \frac{3}{5}BR$, $RM = \frac{3}{5}BM$

Prove:
 $\triangle MGR \sim \triangle MRB$

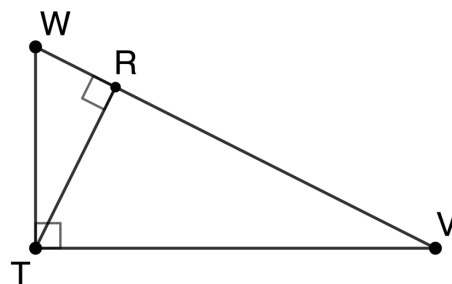


Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.

Proof F

Given: $\overline{WT} \perp \overline{VT}$ and
 $\overline{WV} \perp \overline{RT}$

Prove: $\triangle RTW \sim \triangle TVW$

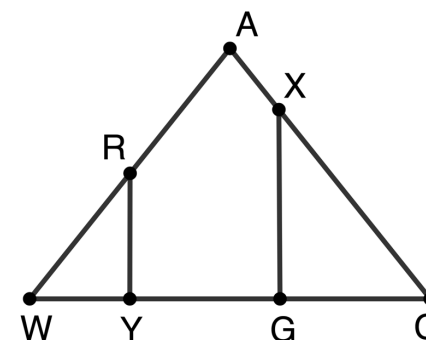


Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7.	7.
8.	8.
9.	9.
10.	10.

Proof G

Given: $\overline{WA} \cong \overline{CA}$,
 $\overline{RY} \perp \overline{WC}$, $\overline{XG} \perp \overline{WC}$

Prove: $\triangle RYW \sim \triangle XGC$



Statement	Reason
1.	1.
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7.	7.
8.	8.
9.	9.

10.	10.
11.	11.
12.	12.
13.	13.
14.	14.



6.9.3 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about proving triangle similarity.

Unit 6, Lesson H3: Justifying Similarity



Benchmark

MA.912.GR.2.9 Justify the criteria for triangle similarity using the definition of similarity in terms of non-rigid transformations.



Learning Target

- ☐ I can use the definition of similarity to justify that SSS similarity, SAS similarity, and AA similarity are sufficient to show that two triangles are similar.



6.H3.1 Warm-Up: Which One Doesn't Belong?

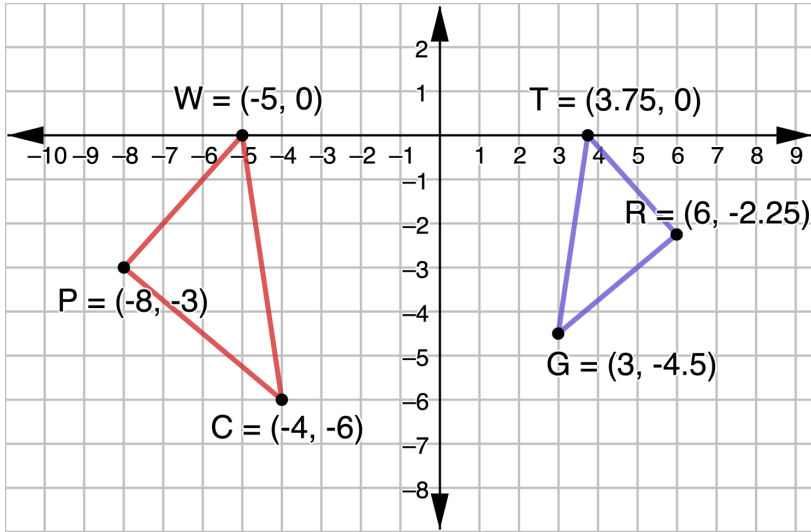
Image A	Image B
Image C	Image D

1. Choose an image that does not belong and explain your choice.



6.H3.2 Collaboration: Justifying Similarity

1. $\triangle PWC$ and $\triangle RTG$ are shown.



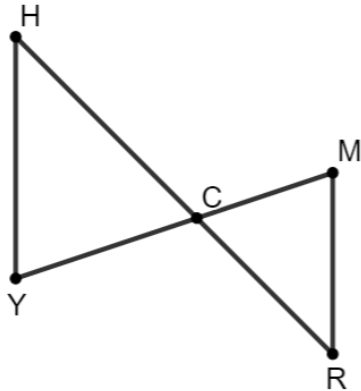
- What do you notice about the two triangles?
- What transformations can be applied so that $\triangle WPC$ is mapped to $\triangle TRG$?
- What can you conclude from the transformations?

Omario's and Ernest's answers to the question "If $\triangle PWC$ and $\triangle RTG$ are similar by angle-angle similarity, what information would need to be given?" are shown.

Omario	Ernest
$\angle WPC \cong \angle TRG$ and $\angle WCP \cong \angle TGR$	$\angle PWC \cong \angle RTG$ and $\angle PCW \cong \angle RGT$

- What do you notice about their answers?
- Explain why both students correctly answered the question.

2. $\triangle MRC$ and $\triangle YHC$ are shown.



- a. Complete the table.

Given Information	True or False? $\triangle MRC \sim \triangle YHC$ by AA Similarity
$\angle HCY \cong \angle RCM$ $\angle H \cong \angle R$	☐ True ☐ False
$\angle HCY \cong \angle RCM$ $\frac{MR}{YH} = \frac{MC}{YC}$	☐ True ☐ False
$\angle H \cong \angle R$ $\angle Y \cong \angle M$	☐ True ☐ False

- b. Select all of the transformations in a sequence that could be applied so that $\triangle MRC$ coincides with $\triangle YHC$:
- ☐ rotate $\triangle MRC$ about point C
 - ☐ rotate $\triangle MRC$ about point R
 - ☐ reflect $\triangle MRC$ across $\overline{MR}$
 - ☐ reflect $\triangle MRC$ across $\overline{YH}$
 - ☐ translate $\triangle MRC$ left and up
 - ☐ dilate $\triangle MRC$ using point C as the center
 - ☐ dilate $\triangle MRC$ using point R as the center

- c. Discuss with your partner what is true about the triangles when all of the transformations have been applied.



Similarity in terms of non-rigid transformations is one or more rigid transformations (reflection, rotation, translation) combined with a dilation that leads to two figures being similar.



Similarity is having exactly the same shape but not necessarily the same size. Two figures are similar if and only if one figure can be obtained from the other by a dilation, or a sequence of transformations that includes a dilation.

- d. Use the definition of similarity in terms of non-rigid motions to justify that $\triangle MRC$ and $\triangle YHC$ are similar.

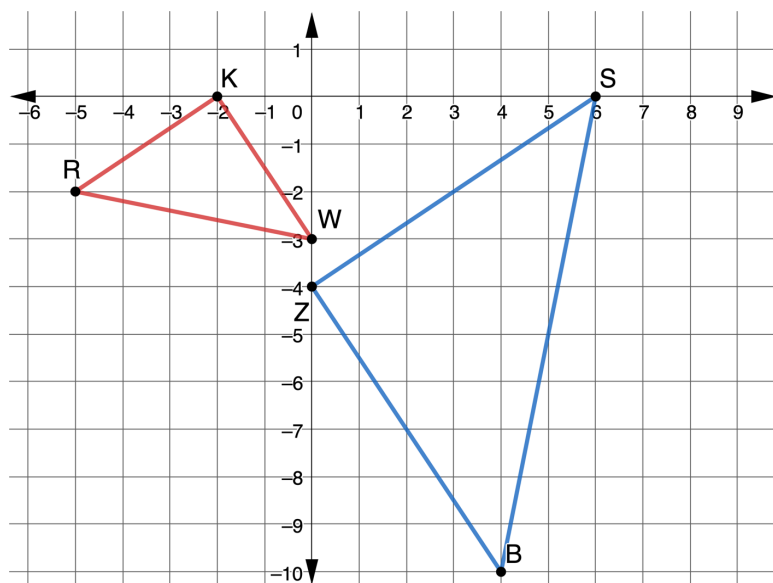
3. Consider the following statement.

If it is possible to find one or more transformations that will transform a figure such that all corresponding points coincide with the other figure, then the figures will be similar.

Explain what this statement means in your own words.

4. $\triangle RKW$ and $\triangle BZS$ are shown.

Assume: $\frac{RW}{BS} = \frac{WK}{SZ} = \frac{KR}{ZB}$



a. Explain how you can use patty paper to verify $\triangle RKW \sim \triangle BZS$.

b. Write a series of transformations that will map $\triangle RKW$ onto $\triangle BZS$ so that $\triangle RKW \sim \triangle BZS$.

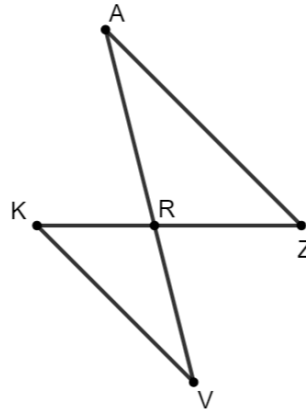
c. If it is given that $\triangle BZS$ is a result of $(x, y) \rightarrow (-y, x)$ then $(x, y) \rightarrow (2x, 2y)$ to $\triangle RKW$, are the triangles congruent?

d. If it is given that $\triangle BZS$ is a result of $(x, y) \rightarrow (-y, x)$ then $(x, y) \rightarrow (2x, 2y)$ to $\triangle RKW$, are the triangles similar?

e. Discuss with your partner what criteria would have to be given so that SAS similarity would be shown as sufficient to justify triangles are similar. Summarize your discussion.

5. Katelyn, Diego, and Amelia are writing conclusions about proving similarity in triangles. They used $\triangle KRV$ and $\triangle ZRA$ to write their conclusions. Their conclusions are shown.

Assume: $\angle VKR \cong \angle AZR$ and
 $\angle VRK \cong \angle ARZ$



Katelyn's Conclusion	Diego's Conclusion	Amelia's Conclusion
To prove similarity of two triangles, apply the definition of congruence in terms of rigid motions to show that two angles that are assumed to be congruent are congruent. This is sufficient to prove similarity of two triangles.	Show that two triangles are congruent using rigid motions and dilation. Then, apply the definition of similarity in terms of non-rigid transformations. This is sufficient to prove the triangles are similar.	By applying a series of rigid transformations and a dilation, it could be shown that the triangles are similar by showing that one set of sides are proportional and assumed two angles are congruent.

- a. Whose conclusion do you agree with?

- b. Explain why you agree with that student.

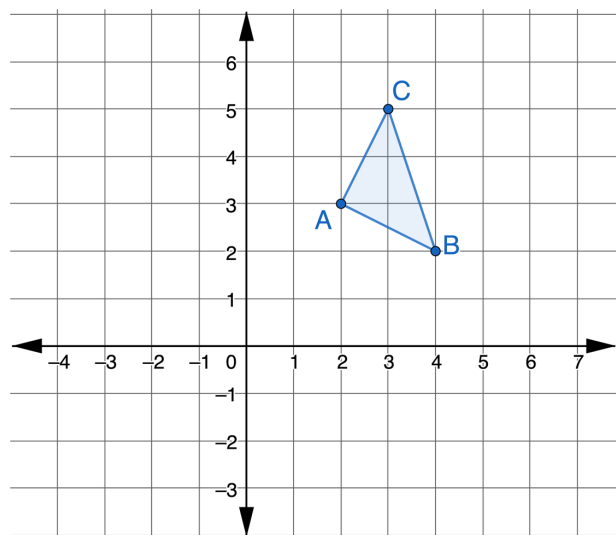
- c. Pick one student whose conclusion is incorrect. Write a note to them to help them understand their error.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills



6.H3.3 Wrap-Up: Cool Down

1. $\triangle ABC$ is shown.



A sequence of three transformations is shown.

- $(x, y) \rightarrow (x - 4, y - 1)$
- $(x, y) \rightarrow (y, x)$
- $(x, y) \rightarrow (1.5x, 1.5y)$

- Draw the result of each transformation.
- Complete the table by noting if the preimage and image are congruent or similar.

Transformation	Relationship of Preimage to Image
$(x, y) \rightarrow (x - 4, y - 1)$	
$(x, y) \rightarrow (y, x)$	
$(x, y) \rightarrow (1.5x, 1.5y)$	

Unit 6, Lesson 10: Similar Figures



Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.



Learning Target

- ☐ I can solve mathematical problems using corresponding angles and sides of similar figures.

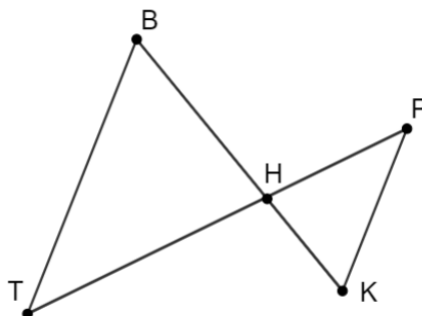


6.10.1 Warm-Up: Bell Work

1. Complete the proof.

Given: $\overline{BT} \parallel \overline{PK}$

Prove: $\triangle BHT \sim \triangle KHP$



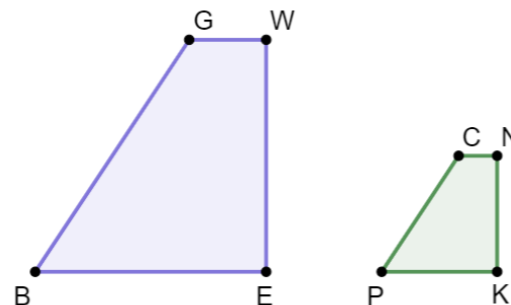
Statement	Reason
1. $\overline{BT} \parallel \overline{PK}$	1.
2.	2.
3.	3.
4. $\triangle BHT \sim \triangle KHP$	4.



6.10.2 Collaboration: Similar Figures

With your partner, answer the following questions.

1. $GWEB$ and $CNKP$ are shown.



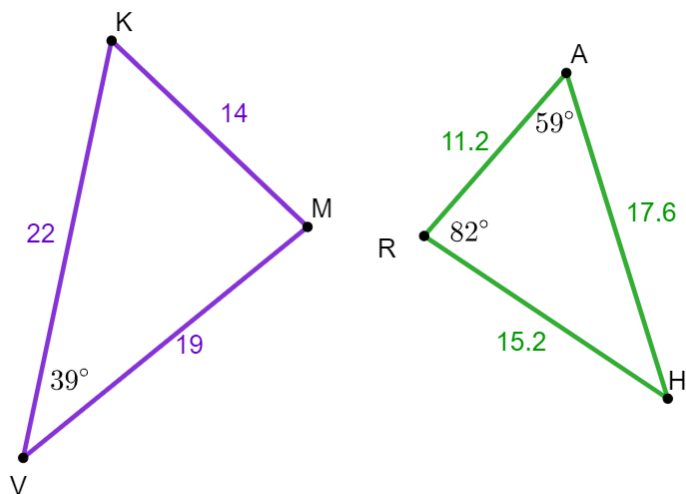
a. If $GWEB \sim CNKP$, what is true about the corresponding angles?

b. If $GWEB \sim CNKP$, what is true about the corresponding sides?

c. Fill in the table given that $GWEB \sim CNKP$.

Corresponding Angles	Corresponding Sides
$\angle GBE \cong \angle$	$\frac{PK}{\quad} = \frac{KN}{\quad}$
$\angle PKN \cong \angle$	$\frac{CN}{\quad} = \frac{EW}{\quad}$
$\angle PCN \cong \angle$	$\frac{GB}{\quad} = \frac{WG}{\quad}$
$\angle GWE \cong \angle$	$\frac{EB}{\quad} = \frac{PC}{\quad}$

2. $\triangle VKM$ and $\triangle HRA$ are shown. $KM = 14$, $MV = 19$, $VK = 22$, $RA = 11.2$, $AH = 17.6$, $HR = 15.2$, $m\angle KVM = 39^\circ$, $m\angle HRA = 82^\circ$, and $m\angle RAH = 59^\circ$.



- d. Complete the table.

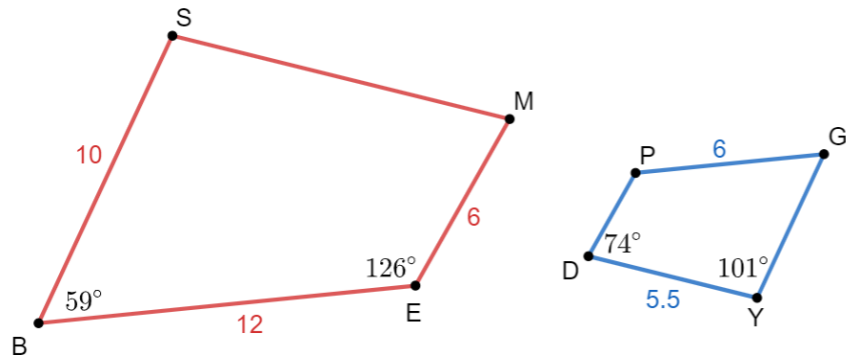
Angle Measures	Side Proportions
$m\angle VKM =$	$\frac{MK}{RA} =$
$m\angle KVM =$	$\frac{MV}{RH} =$
$m\angle RHA =$	$\frac{KV}{AH} =$

- e. What scale factor are the sides of $\triangle VKM$ multiplied by to find the corresponding sides of $\triangle HRA$?

- f. If $MK = 21$, what should the length of $\overline{RA}$ be so that the triangles are similar?

- a. Using the given information, what can you conclude about $\triangle VKM$ and $\triangle HRA$?
- b. What similarity theorem or postulate could be used to prove the triangles are similar?
- c. Write a similarity statement for $\triangle VKM$ and $\triangle HRA$.

3. $SMEB$ and $YDPG$ are shown. $SB = 10$, $BE = 12$, $EM = 6$, $DY = 5.5$, $PG = 6$, $m\angle SBE = 59^\circ$, $m\angle BEM = 126^\circ$, $m\angle PDY = 74^\circ$, and $m\angle DYG = 101^\circ$.



- a. If $SMEB \sim YDPG$, determine the measure of each.

$$m\angle BSM = \underline{\hspace{2cm}} \qquad m\angle SME = \underline{\hspace{2cm}}$$

$$m\angle DPG = \underline{\hspace{2cm}} \qquad m\angle PGY = \underline{\hspace{2cm}}$$

$$SM = \underline{\hspace{2cm}} \qquad PD = \underline{\hspace{2cm}}$$

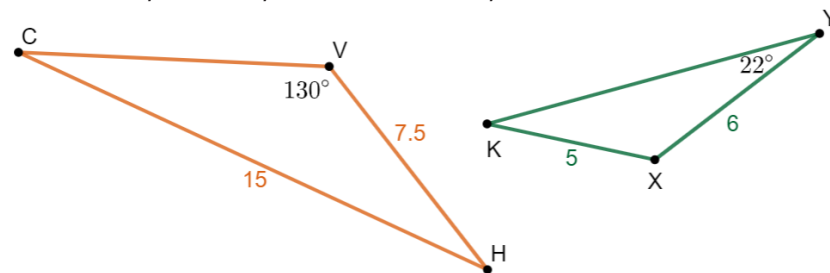
$$GY = \underline{\hspace{2cm}}$$

- b. Explain how the similarity relationship, $SMEB \sim YDPG$, helped determine which sides of the quadrilaterals are proportional.



6.10.3 Your Turn!

1. $\triangle CVH$ and $\triangle YXK$ are shown where $VH = 7.5$, $CH = 15$, $KX = 5$, $XY = 6$, $m\angle CVH = 130^\circ$, and $m\angle XYK = 22^\circ$.



- a. Given that $\triangle CVH \sim \triangle YXK$, determine the measure of each angle.

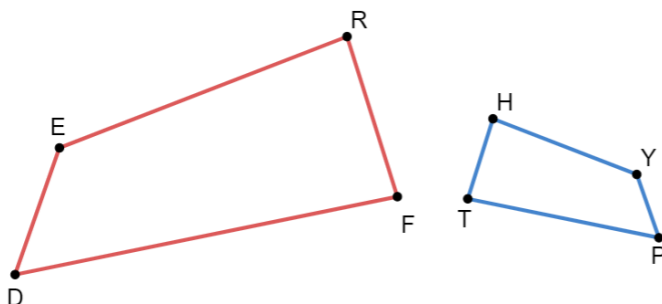
$$m\angle VCH = \underline{\hspace{2cm}} \qquad m\angle VHC = \underline{\hspace{2cm}}$$

$$m\angle YKX = \underline{\hspace{2cm}} \qquad m\angle YXK = \underline{\hspace{2cm}}$$

- b. Given that $\triangle CVH \sim \triangle YXK$, determine the measure of each side.

$$CV = \underline{\hspace{2cm}} \qquad KY = \underline{\hspace{2cm}}$$

2. $RFDE$ and $HTPY$ are shown, where $RE = 6.4$, $ED = 2.8$, $HY = 3.2$, $HT = 1.7$, $PT = 4$, $m\angle ERF = 86^\circ$, $m\angle EDF = 59^\circ$, and $m\angle HYP = 84^\circ$.



Assume: $RFDE \sim HTPY$

- a. Find the measure of $\angle DER$. Show your work or explain your thinking.

- b. Determine each measure.

$$m\angle HTP = \underline{\hspace{2cm}} \quad m\angle DFR = \underline{\hspace{2cm}}$$

$$m\angle THY = \underline{\hspace{2cm}} \quad m\angle YPT = \underline{\hspace{2cm}}$$

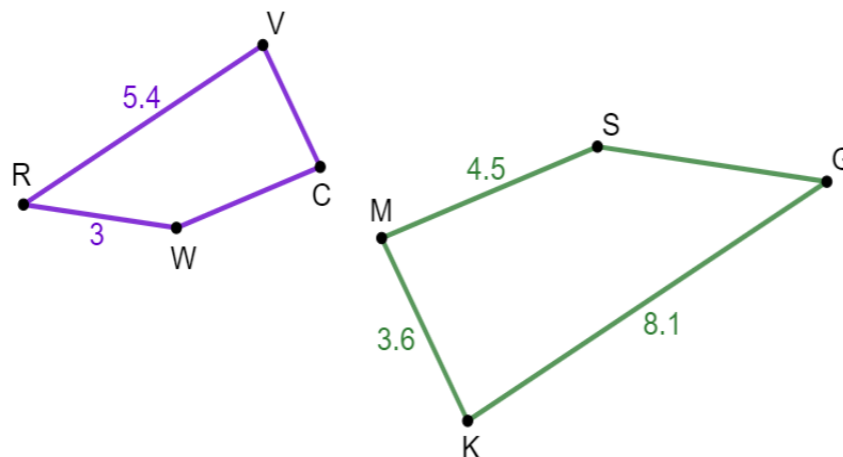
- c. What is the ratio of the corresponding sides of $RFDE$ to $HTPY$?

- d. Determine each measure.

$$DF = \underline{\hspace{2cm}} \quad RF = \underline{\hspace{2cm}}$$

$$YP = \underline{\hspace{2cm}}$$

3. Kirstyn and Roberto were finding the missing side measures for $RVCW$ and $GKMS$. They were given $RVCW \sim GKMS$, $VR = 5.4$, $RW = 3$, $SM = 4.5$, $MK = 3.6$, and $KG = 8.1$.



Kirstyn's and Roberto's work to find the measure of $\overline{VC}$ is shown.

Kirstyn's Work	Roberto's Work
$\frac{8.1}{5.4} = 1.5$	$\frac{8.1}{5.4} = 1.5$
$3.6 \times 1.5 = 5.4$	$3.6 \div 1.5 = 2.4$
$VC = 5.4$	$VC = 2.4$

- a. Who do you agree with?
- b. Explain the error in the incorrect work.
- c. Complete the statements.
- $$CW = \underline{\hspace{2cm}} \quad SG = \underline{\hspace{2cm}}$$



6.10.4 Wrap-Up: Reflection

1. One thing I struggled with today was ...
2. One thing I need to practice more is ...
3. One thing I understand very well is ...



Unit 6, Lesson 11: Solving Problems Using Similarity – Part 1



Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.



Learning Target

- ☐ I can solve mathematical problems involving similarity in two-dimensional figures.



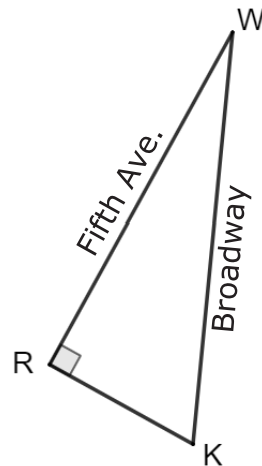
6.11.1 Warm-Up: The Flatiron Building



The Flatiron Building, designed by Daniel Burnham and built in 1902, is a triangular 22-story building located in New York City. The building sits on a triangular block formed by 5th Avenue, Broadway, and East 22nd Street. The building got its name from an old-fashioned flat iron.



Although the ground base appears to be an isosceles triangle, the base of the Flatiron Building is actually a right triangle. Morgan used a map to draw a triangle that models the base of the building.

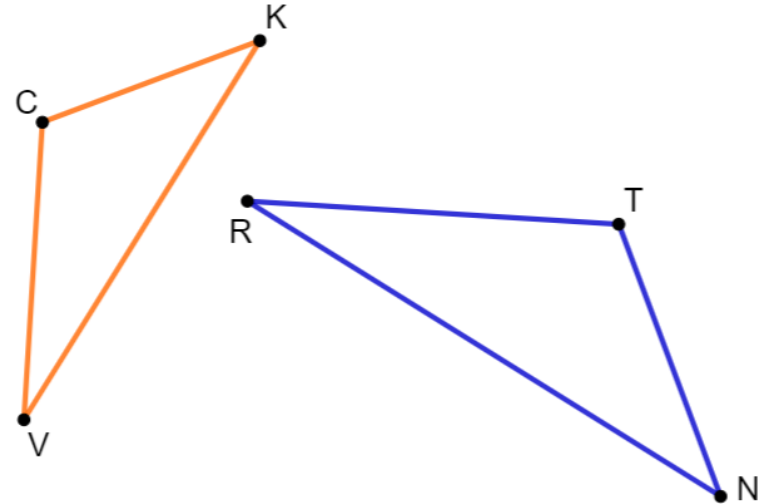


1. What other measurement should Morgan check to be sure her drawing is similar to the building?



6.11.2 Collaboration: Similar Figures

1. $\triangle CKV$ and $\triangle TNR$ are shown, where $\triangle CKV \sim \triangle TNR$.



- a. Some measurements of the angles in $\triangle CKV$ and $\triangle TNR$ are given.
 $m\angle TRN = 28^\circ$, $m\angle VCK = (8x + 26)^\circ$, and
 $m\angle RTN = (12x - 8)^\circ$.

Determine each.

$$x = \underline{\hspace{2cm}} \qquad m\angle CKV = \underline{\hspace{2cm}}$$

Some measurements of the sides of $\triangle CKV$ and $\triangle TNR$ are given.

$$VK = 8, CK = 5z + 1.7, CV = 5.4, TR = 2y - 1.25, \\ RN = 10, \text{ and } TN = 5.25.$$

b. What is the scale factor from $\triangle CKV$ to $\triangle TNR$?

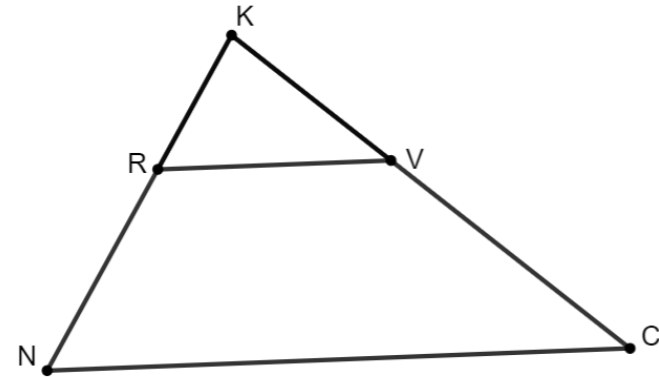
c. Determine the value of each variable.

$$z = \underline{\hspace{2cm}} \qquad y = \underline{\hspace{2cm}}$$

d. Determine the measure of each side.

$$CK = \underline{\hspace{2cm}} \qquad TR = \underline{\hspace{2cm}}$$

2. $\triangle KVR$ and $\triangle KCN$ are shown, where $\triangle KVR \sim \triangle KCN$.



Some measurements of the sides are given.

$$RV = 4.6, KR = 3, KV = 5x - 11, NC = 11.5, \text{ and } \\ KC = 3x + 1.$$

Maria used the information to determine the length of $\overline{KC}$. Her work is shown.

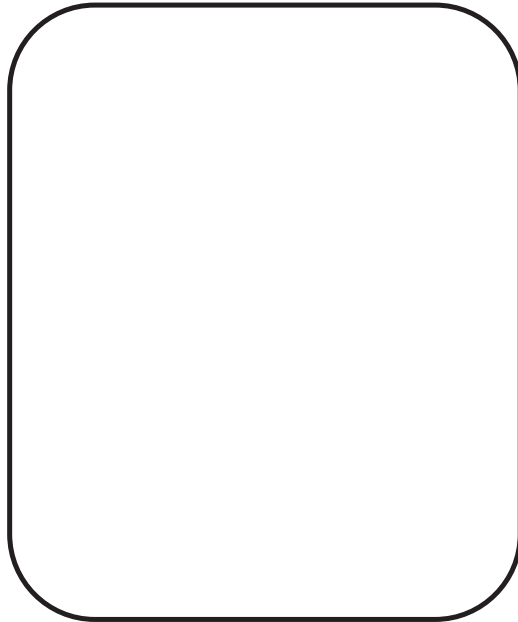
Maria's Work	
First, Maria found the scale factor.	$\frac{NC}{RV} = \frac{11.5}{4.6} = 2.5$
Second, she used the scale factor to write an equation. Then she solved for x .	$\begin{aligned} 5x - 11 &= 2.5(3x + 1) \\ 5x - 11 &= 7.5x + 2.5 \\ -2.5x &= -13.5 \\ x &= -5.4 \end{aligned}$
Lastly, she substituted x into an equation.	$\begin{aligned} KC &= 3(-5.4) + 1 \\ KC &= -15.2 \end{aligned}$

Maria knows side lengths cannot be negative values, but she does not know what she did wrong.

a. Find and circle Maria's error.

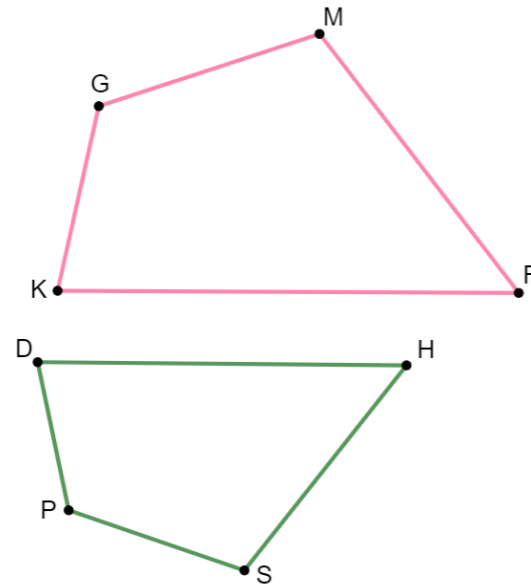
b. Write a note to Maria explaining what her mistake was and how she can learn from the mistake.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills



6.11.3 Your Turn!

1. $GMRK$ and $PSHD$ are shown, where $GMRK \sim PSHD$.



Some measurements of the sides of $GMRK$ and $PSHD$ are given.

$KG = 12$, $GM = 4x - 12$, $MR = 7k - 15$, $DP = 9.6$, $SP = 12.8$, $KR = 30$, and $SH = 4k - 4$.

a. Determine the value of each variable.

$x =$ _____

$k =$ _____

- b. Determine the value of each side.

$$GM = \underline{\hspace{2cm}} \quad DH = \underline{\hspace{2cm}}$$

$$SH = \underline{\hspace{2cm}}$$

- c. What is the perimeter of $GMRK$?

Some measurements of the angles of $GMRK$ and $PSHD$ are given.

$$m\angle KGM = (11w - 12)^\circ, m\angle GMR = 110^\circ, \\ m\angle GKR = (6z + 18)^\circ, m\angle DPS = (8w + 24)^\circ, \text{ and} \\ m\angle PDH = (8z - 2)^\circ.$$

- d. Determine the value of each variable.

$$w = \underline{\hspace{2cm}} \quad z = \underline{\hspace{2cm}}$$

- e. Determine the value of each angle.

$$m\angle KGM = \underline{\hspace{2cm}} \quad m\angle DHS = \underline{\hspace{2cm}}$$

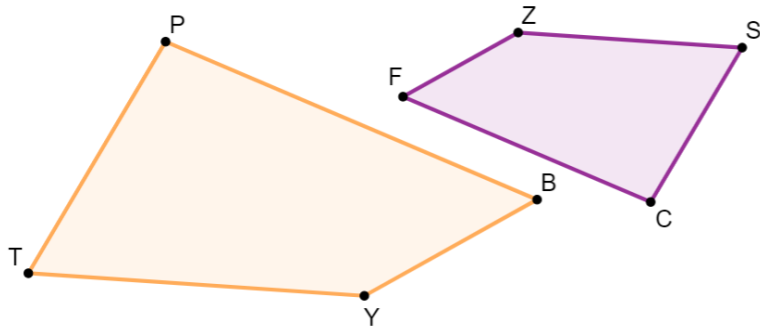
$$m\angle PDH = \underline{\hspace{2cm}}$$





6.11.4 Wrap-Up: Error Analysis

1. Cheryl and Austin are finding the measure of side SC . They were given the following information about similar quadrilaterals $ZSCF$ and $YTPB$ that are shown. $\triangle ZSCF \sim \triangle YTPB$, $ZS = 15$, $SC = 5x - 8$, $PT = 3x + 6$, and $YT = 22.5$.



Cheryl's and Austin's equations are shown.

Cheryl's Work	Austin's Work
$15(5x - 8) = 22.5(3x + 6)$	$1.5(5x - 8) = 3x + 6$

- Determine whose equation, if any, is set up incorrectly. Show your work or write an explanation.
- What is the length of $\overline{SC}$?

Unit 6, Lesson 12: Solving Problems Using Similarity – Part 2



Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.



Learning Target

- ☐ I can solve real-world problems involving similarity in two-dimensional figures.



6.12.1 Warm-Up: Biomedical Career

Liz Williams and Kate Lomas co-founded the Australian-based biomedical technology company, Hemideina. In Kate's studies she found that the insect wētā has an amazing hearing system. Through her studies she invented an implant that can help those with hearing loss live a life without limits.



1. List at least one advantage that Kate may have had as an artist that helped her create her invention.

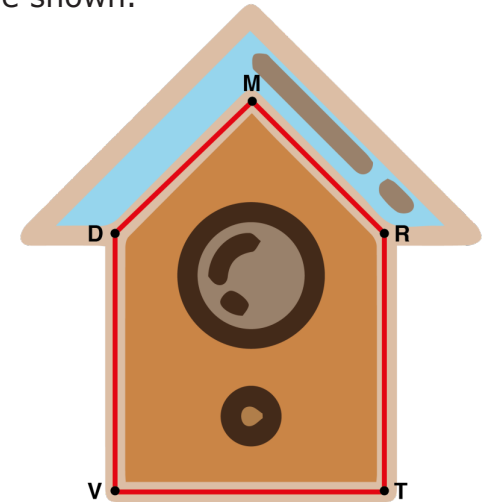
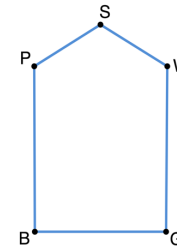
Both Liz and Kate express excitement knowing that their skills in science can be used to solve real-world problems.

2. What is one real-world problem that you think creativity, math, and science could be used to solve?



6.12.2 Guided Instruction: Similar Figures in the Real World

1. Haley built a birdhouse by first drawing a blueprint. She then constructed the birdhouse using the dimensions from the blueprint. Her blueprint, $PSWGB$, and birdhouse, $DMRTV$, are shown.



To build the birdhouse, Haley multiplied each of the dimensions from the blueprint by 4 to get the dimensions of the face of the birdhouse. She used equations to represent the sides.

$PS = (y + 1.2)$ inches (in.), $PB = (3x - 1.5)$ in.,
 $BG = (2w - 6.4)$ in., $VT = (4w - 5.6)$ in., $DV = (4x + 10)$ in.,
 and $DM = (6.2y + 2.6)$ in.

- a. What is the value of x ?

b. What is the length of $\overline{DV}$? Include units.

c. What is the value of y ?

d. What is the length of $\overline{PS}$? Include units.

e. What is the value of w ?

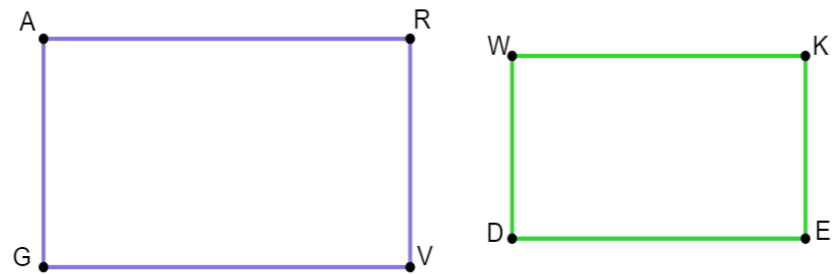
f. What is the length of $\overline{VT}$? Include units.

g. If $DV = RT$ and $DM = MR$, what is the perimeter, in inches, of the face of the birdhouse?



6.12.3 Guided Practice: Similar Figures in the Real World

1. Moving-R-Us creates boxes in the shape of a rectangular prism. The company uses a computer program to calculate similar size boxes by using equations. The diagram shows the rectangular bases, $ARVG$ and $WKED$, of two different size boxes, where $WD = 2$ inches (in.), $AR = (7w - 30)$ in., $RV = 2.5$ in., and $WK = (6w - 26)$ in.



a. What is the value of w ?

b. What is the area of $ARVG$? Include units.

- c. What is the perimeter of $WKED$? Include units.

Moving-R-Us created a third similar rectangular box. The base, $MHST$, is shown, where $MH = 6.25$ in.

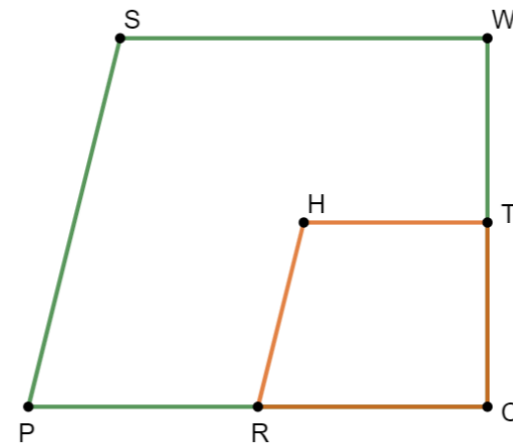


- d. What is the length of $\overline{MT}$? Include units.
- e. What is the perimeter, in inches, of $MHST$?



6.12.4 Try It: Solving in Real-World Context

1. Gloria, a landscape engineer, created a trapezoidal-shaped garden with a similar trapezoidal-shaped garden inside. A diagram of the gardens, $SWCP$ and $HTCR$, are shown, where $PC = (3y + 4)$ feet (ft), $RC = (y + 3)$ ft, $SP = 8.2$ ft, $SW = (7x - 13)$ ft, $HT = (2x - 2)$ ft, and $HR = 4.1$ ft.



- a. What is the value of x ?
- b. What is the length of side SW ? Include units.
- c. What is the value of y ?

- d. What is the length of side RC ? Include units.
- e. If $SW = WC$, what is the perimeter, in feet, of the larger garden?

After Gloria created the gardens, she decided to check the angle measures of each corner. She had the following information written down: $m\angle SWC = (12z + 6)^\circ$, $m\angle HTC = (7z + 41)^\circ$, $m\angle TCR = 90^\circ$, $m\angle SPC = (5w + 11)^\circ$, and $m\angle HRC = (4w + 24)^\circ$.

- f. What is the value of z ?
- g. What is the value of w ?
- h. What is the measure of $\angle RHT$?



6.12.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to let them know what they missed in today's lesson.

Unit 6, Lesson 13: Similarity in the Real World



Benchmark

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.



Learning Target

- ☐ I can solve real-world problems involving similarity in two-dimensional figures.



6.13.1 Warm-Up: With Math I CAN

1. What struggles have you had in mathematics?
2. Students often ask teachers, “When will I ever use this?” What is one thing in mathematics you’ve wondered if you would ever use?
3. Sometimes it isn’t that a job requires you to prove a theorem, but to use the logical thinking you learned in proving a theorem. Describe a few careers that you’re interested in and what can you do with math.



6.13.2 Activity: Real-World Problem Gallery Walk

Complete each problem by recording your work and answer in the spaces provided.

Problem	Work	Answer
Problem A		
Problem B		

Problem C		
Problem D		



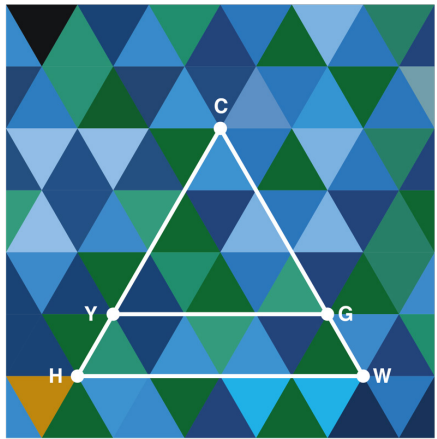
Problem E		
Problem F		



6.13.3 Wrap-Up: Ticket Out the Door

- Beth and Trevor are trying to find the lengths of the triangles on the quilt.

Beth and Trevor know that $\triangle CYG \sim \triangle CHW$, $CG = 21$ inches (in), $YG = (2.5x + 1.5)$ in, $GW = 7$ in, and $HW = (6x - 14)$ in. Their work to find the scale factor is shown.



Beth's Work	Trevor's Work
$\frac{CG}{GW} = \frac{21}{7}$ Scale Factor is 3	$\frac{CG}{CW} = \frac{21}{28}$ Scale Factor is 0.75

- Whose work do you agree with?
- What is the length of $\overline{HW}$?